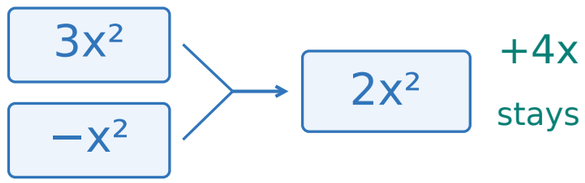
Algebra 1

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 1.1

Collect like terms



$ax^n + bx^n = (a+b)x^n$

Remember: Only combine matching variables AND powers.

02 / 1.2

Multiply every term by every term

	x	4
2x	2x²	8x
3	3x	12

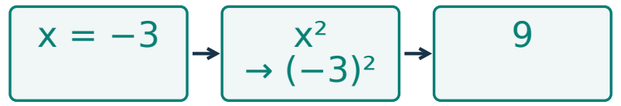
$(2x+3)(x+4) \rightarrow 2x^2+11x+12$

$a(b+c)=ab+ac$; $x^m x^n = x^{m+n}$. m terms × n terms → mn products.

Remember: Index rule here uses non-negative integer powers.

03 / 1.3

Substitute a signed value



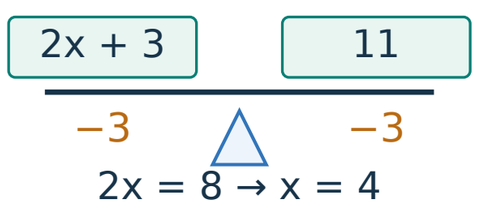
Bracket the negative value before squaring.

Replace each variable by its whole signed value in brackets. Denominator ≠ 0.

Remember: $(-3)^2 = 9$, but $-3^2 = -9$.

04 / 1.4

Keep an equation balanced

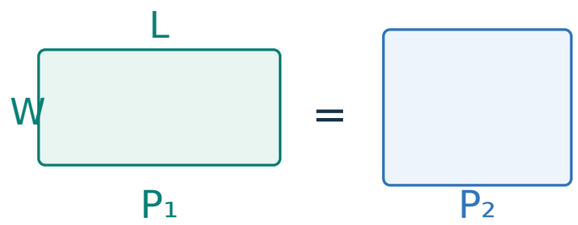


Do the same valid operation to BOTH sides; clear every denominator. Never divide by zero.

Remember: A multiplier applies to every term on both sides.

05 / 1.5

Match two perimeters



Rectangle $P=2L+2W$. Set $P_1=P_2$ when the perimeters are equal.

Remember: Equal boundary lengths do not imply equal areas.

Algebra 1

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

06 / 1.5

Subtract a removed area

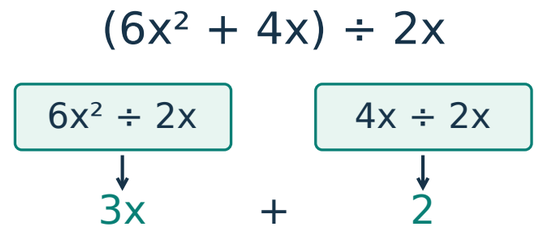


Unshaded area = whole area − shaded area.

Remember: Use square units; a minus before brackets changes every sign.

07 / 1.6

Divide every term

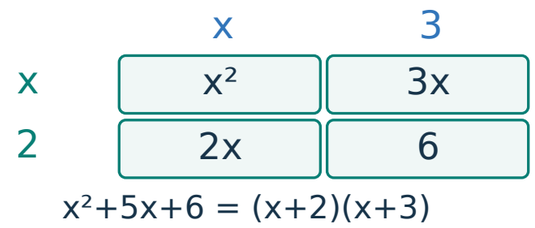


$(A+B-C)/d = A/d + B/d - C/d$; $x^m/x^n = x^{m-n}$.

Remember: Require $d \neq 0$ and $x \neq 0$ for the index division rule.

08 / 1.6

Reverse a multiplication grid

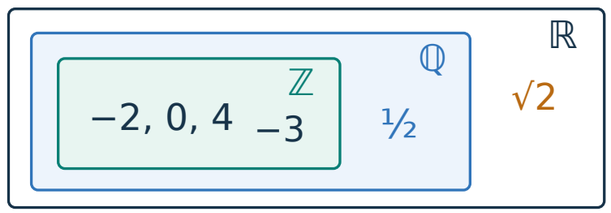


Dividend = divisor × quotient + remainder.

Remember: Divisor ≠ 0; leave spaces for missing powers.

09 / 1.7

Locate a number family



$\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$; rational numbers are p/q with integers p,q and q≠0.

Remember: Recurring decimals are rational; √2 is irrational.

10 / 1.8

Solve and shade an inequality

$-2x \geq -6 \rightarrow x \leq 3$



Filled endpoint: 3 is included.

Multiply or divide by a negative → reverse <, >, ≤ or ≥.

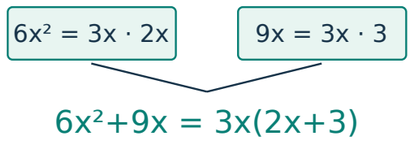
Remember: Adding or subtracting a negative does not reverse the sign.

Factors

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 2.1

Take out the highest common factor



HCF = numerical HCF × lowest power common to every term.

Remember: Check the remaining bracket for further factors.

02 / 2.2

Group to make matching brackets

	p	q
a	ap	aq
b	bp	bq

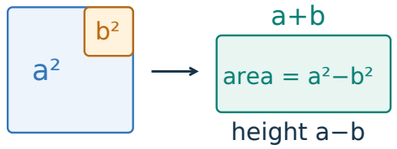
Both rows contain the same bracket: (p+q).

ap+aq+bp+bq=(a+b)(p+q).

Remember: Taking out a negative factor reverses signs inside its bracket.

03 / 2.3

Recognise a difference of squares



a²−b²=(a−b)(a+b).

Remember: Two squares separated by subtraction; not a²+b² over the reals.

04 / 2.4

Find two numbers for a monic quadratic

	x	3
x	x²	3x
2	2x	6

p=2, q=3: sum 5; product 6.

x²+bx+c=(x+p)(x+q): p+q=b, pq=c. x²±2ax+a²=(x±a)².

Remember: Check BOTH the sum and the product.

05 / 2.4

Split the middle term when a ≠ 1

	x	3
2x	2x²	6x
1	x	3

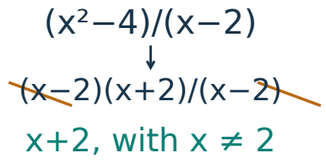
2x²+7x+3: split 7x into 6x+x.

ax²+bx+c: find m+n=b and mn=ac; split bx, then group.

Remember: Use ac for the product, rather than c alone.

06 / 2.5

Cancel factors and keep restrictions

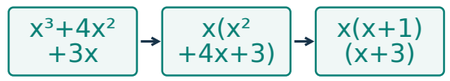


N/D: factor first, cancel common factors only where original D≠0.

Remember: A cancelled denominator factor still creates an excluded input.

07 / 2.5

Factor a cubic before cancelling



First extract x; then factor the quadratic.

Use HCF or grouping → linear factor × quadratic → further factors.

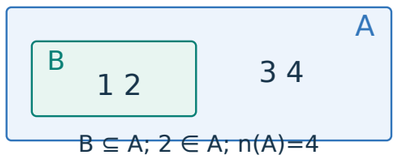
Remember: Cancel complete factors, while retaining original exclusions.

Sets

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 3.1

Read membership, subset and size

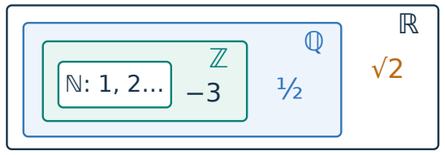


a∈A: member; B⊆A: every B member is in A; n(A): number of distinct members.

Remember: {0} has one member; ∅ has none.

02 / 3.1

Place numbers in nested families

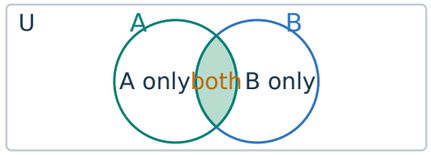


ℕ ⊂ ℤ ⊂ ℚ ⊂ ℝ; irrational = real but not rational.

Remember: Use the convention given for whether ℕ includes 0.

03 / 3.1

Choose union, intersection or complement

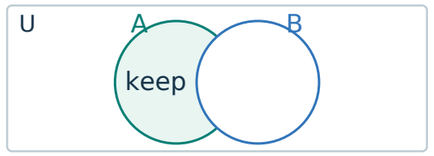


∪: either; ∩: both; A′=U\A; n(A′)=n(U)−n(A). ∪ and ∩ are commutative and associative.

Remember: Union includes overlap. The complement depends on U.

04 / 3.2

Find A but not B

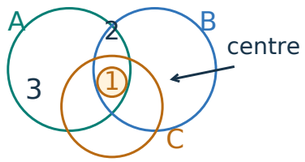


A\B = A∩B′.

Remember: Difference is generally neither commutative nor associative.

05 / 3.3

Fill a three-set diagram from the centre

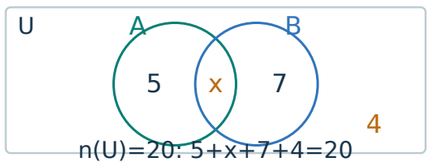


AB-only=n(A∩B)−n(A∩B∩C). n(A∪B∪C)=n(A)+n(B)+n(C)−n(A∩B)−n(A∩C)−n(B∩C)+n(A∩B∩C).

Remember: Pair intersections include the centre unless “only” is specified.

06 / 3.4

Find a missing region or total



Disjoint regions add once. Union = n(U) − outside.

Remember: Counts of people must be non-negative integers.

07 / 3.4

Find the smallest or largest overlap

N=20, a=12, b=11



overlap can range from 3 to 11

max(0,a+b−N) ≤ overlap ≤ min(a,b); union=a+b−overlap.

Remember: a and b are set sizes inside a universe of size N.

08 / 3.4

Distinguish people from memberships

1 1 1

1 set 2 sets 3 sets

3 people → 1+2+3 = 6 memberships

Union=$E_1+E_2+E_3$. Membership total=$E_1+2E_2+3E_3$.

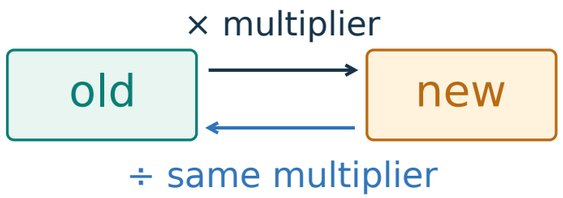
Remember: E_k counts people in exactly k sets; apply ratios to named regions.

Applied Arithmetic

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 4.1

Move forwards or backwards through a percentage

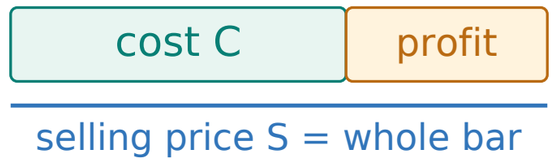


New=old(1±r/100); old=new/(1±r/100).

Remember: Reverse using the same non-zero multiplier.

02 / 4.1

Choose the correct profit percentage base



Profit=S−C; markup%=100×profit/C; margin%=100×profit/S; loss%=100×loss/C.

Remember: C=cost, S=selling price. Markup uses C; margin uses S.

03 / 4.2

Charge each usage band separately



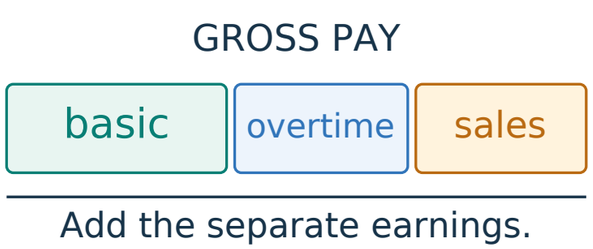
Add the bands + fixed charge.

Subtotal = fixed charge + Σ(band units × band rate).

Remember: Only the units inside a band receive that band’s rate.

04 / 4.3

Build gross pay from its parts



Gross pay = basic pay + overtime pay + commission.

Remember: Apply each hours or sales rate only within its own band.

05 / 4.3

Apply supplied tax bands and credits



Subtract all deductions from gross pay.

Gross tax=Σ(band income×rate); income tax=max(0,gross tax−credits); net=gross−all deductions.

Remember: Calculate USC and PRSI separately using the question’s rules.

06 / 4.3

Work backwards from net pay or tax paid

gross − deductions = net



Solve, then check the tax band.

Within one band: write net as a function of gross. Fully used credit=gross tax−income tax paid.

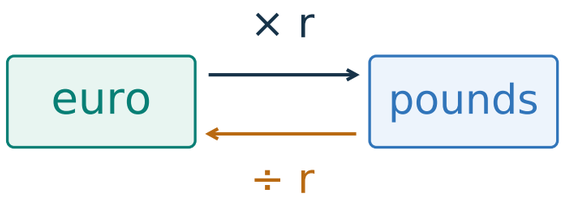
Remember: At zero income tax, an unused larger credit cannot be determined uniquely.

Applied Arithmetic

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 4.4

Convert currency in the right direction

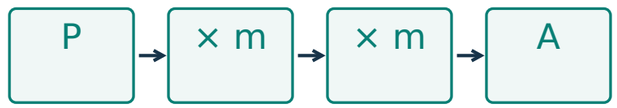


If €1=£r: pounds=euro×r; euro=pounds/r, with r>0.

Remember: Apply fees at the stated stage and in the stated currency.

08 / 4.5

Repeat a growth or depreciation multiplier



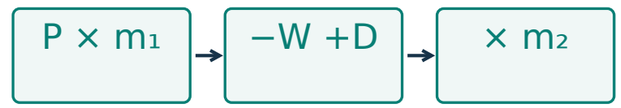
Repeat the multiplier n times.

$A=P(1+r/100)^n$; depreciation $A=P(1-r/100)^n$.

Remember: Match rate and period; apply each change to the current amount.

09 / 4.5

Put rate changes and transactions in order



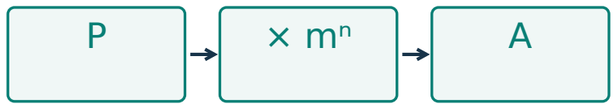
balance 1 → transaction → balance 2

$A=(P\,m_1-W+D)m_2$.

Remember: W and D occur after period 1 here; follow the actual timing.

10 / 4.5

Recover the principal or repeated rate



Reverse with division or an nth root.

$P=A/(m_1m_2\ldots)$; equal periods: $m=(A/P)^{(1/n)}$. Growth $r=100(m-1)\%$; depreciation $r=100(1-m)\%$.

Remember: Use positive balances; distinguish a multiplier from a percentage.

11 / 4.5

Tax the interest, not the principal



Keep interest × (1−t); principal stays.

Net interest=P×r×(1−t), with decimal rates r and t.

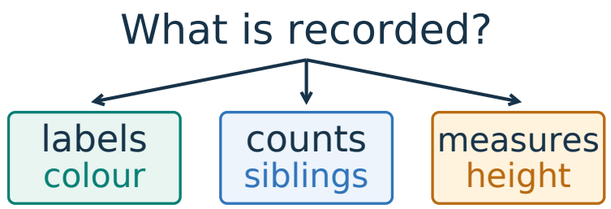
Remember: Interest tax applies to interest; retain the principal separately.

Statistics 1 - Collecting data

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 5.1

Choose the variable and data type

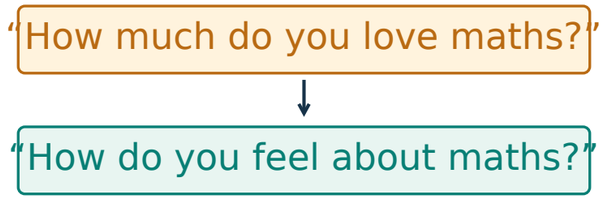


Population: target group. Variable: recorded feature. Categorical=labels; discrete=counts; continuous=measurements.

Remember: Rounding a measurement does not turn it into a count variable.

02 / 5.1

Repair a biased questionnaire

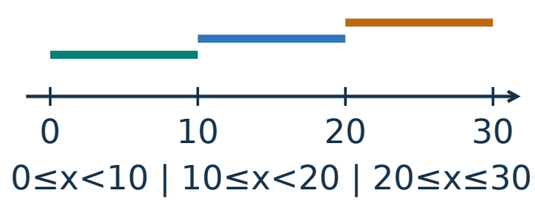


Use neutral wording, one idea at a time, and a precise reference period.

Remember: Avoid leading language and questions that ask two things at once.

03 / 5.1

Give every response exactly one home

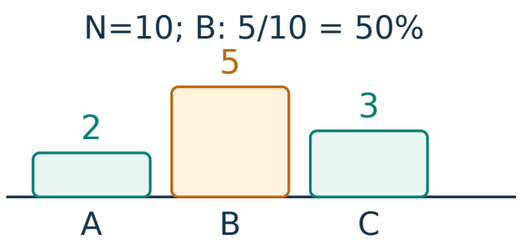


Response categories must be exhaustive and mutually exclusive.

Remember: Use inequalities or clear boundaries so endpoints belong once.

04 / 5.1

Turn counts into relative frequencies

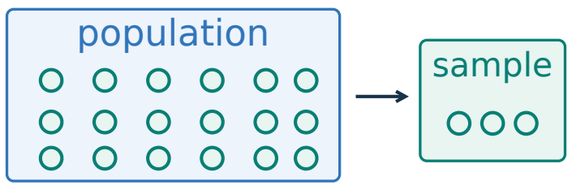


Relative frequency=f/N; percentage=100f/N.

Remember: N counts observations, not categories.

05 / 5.2

Check the source and sampling method

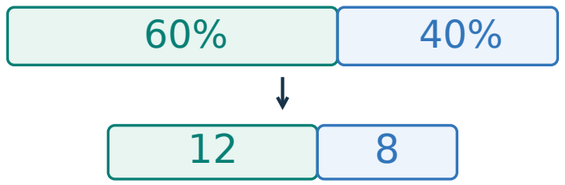


Primary: collected for this investigation. Secondary: already collected. Judge selection and coverage, not just size.

Remember: A large biased sample still describes the wrong group.

06 / 5.2

Take a proportional sample



Group sample = total sample × group population / total population.

Remember: After allocating quotas, select fairly within each group.

Perimeter - Area - Volume

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 6.1

Split a composite shape into familiar pieces



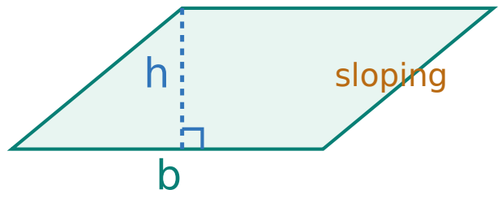
Area: add pieces. Perimeter: trace outside.

Rectangle: A=lw, P=2(l+w). Trapezium: A=(a+b)h/2. Kite/rhombus: A=d_1d_2/2. Add/subtract non-overlapping areas.

Remember: Trace only the outer boundary; h and the two diagonals must be perpendicular as required.

02 / 6.2

Use the perpendicular height



Parallelogram A=bh; triangle A=bh/2.

Remember: h meets the base at 90°; a sloping side is not the height.

03 / 6.3

Choose a circular area or boundary formula

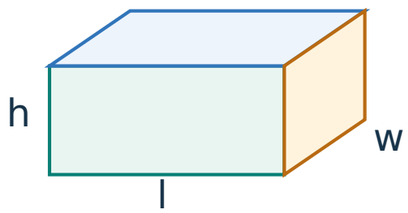


C=2πr; A=πr². Arc=(θ/360)2πr; sector A=(θ/360)πr²; sector P=2r+arc. Semicircle A=πr²/2, P=πr+2r.

Remember: Sector 0<θ<360°. Count straight radii/diameters only when exposed.

04 / 6.4

Find cuboid volume or covering area

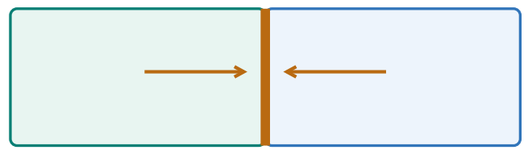


V=lwh; closed SA=2(lw+lh+wh). 1 cm³=1 mL; 1000 cm³=1 L.

Remember: Volume fills the box; surface area covers its faces.

05 / 6.4

Remove hidden faces when joining solids



two touching faces become internal

Exposed SA=SA_1+SA_2−2A_shared; volumes add when solids do not overlap.

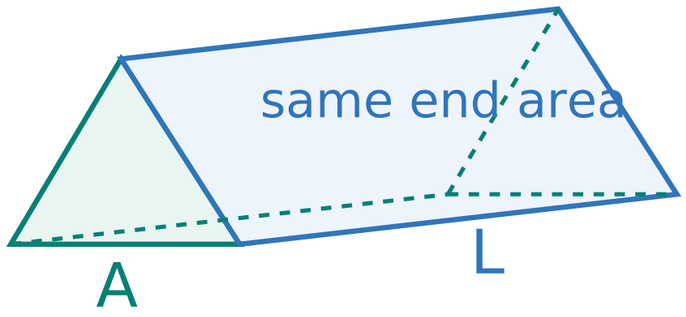
Remember: One full shared face is hidden on BOTH solids.

Perimeter - Area - Volume

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

06 / 6.5

Multiply cross-section area by length

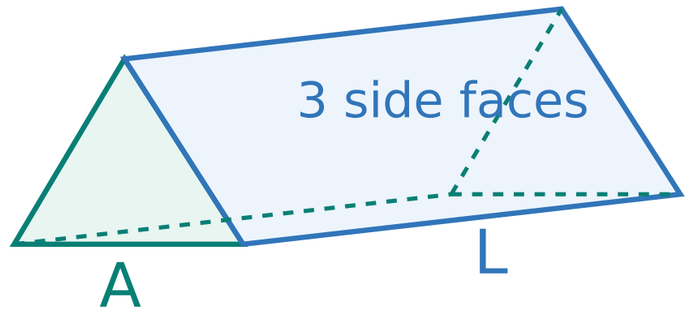


V=A_cross×L; L=V/A_cross, where A_cross>0.

Remember: Find the end area first; a triangular end needs the factor ½.

07 / 6.5

Cover a closed right prism

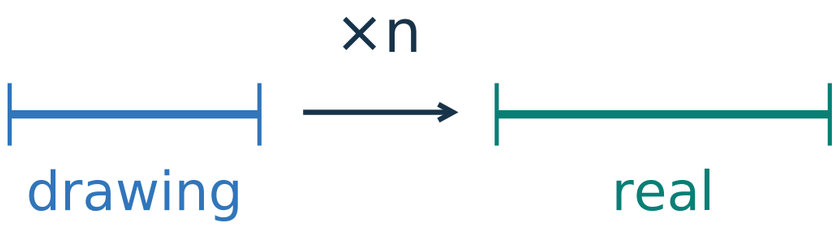


SA=2A_cross+P_cross×L.

Remember: For a closed right prism: two ends plus all side rectangles.

08 / 6.6

Convert a map length both ways



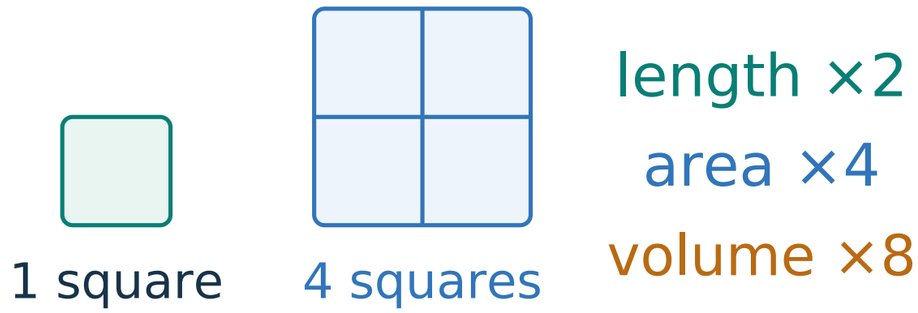
Reverse the arrow by dividing by n.

At scale 1:n: real=n×drawing; drawing=real/n.

Remember: Convert to the same units before applying the ratio.

09 / 6.6

Scale lengths, areas, volumes and costs



Length factor k → area factor k², volume factor k³. Cost=quantity×unit price.

Remember: Use square or cubic factors for square or cubic units.

Statistics 2 - Averages and Variability

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 7.1

Find the mode and ordered median



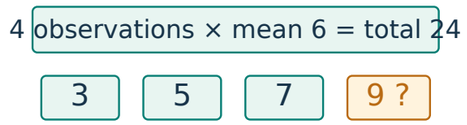
middle = 4; most frequent = 4

Mode=most frequent. Odd N: position (N+1)/2. Even N: mean of positions N/2 and N/2+1.

Remember: Sort first. There may be several modes or no unique mode.

02 / 7.2

Recover a total or missing observation



Mean=total/N; total=N×mean; missing=required total−known total.

Remember: N includes the missing observations.

03 / 7.2

Combine group means with their sizes



(3×8 + 2×13)/5 = 10

Combined mean=Σ(n×group mean)/Σn.

Remember: Weight each mean by its group size.

04 / 7.3

Choose an average that answers the question



An extreme pulls the mean towards it.

Mean: all values. Median: middle, resistant to extremes. Mode: most common value or category.

Remember: Justify the choice in context; no average is always best.

05 / 7.4

Weight values by their frequencies

x	f	fx
2	3	6
5	2	10

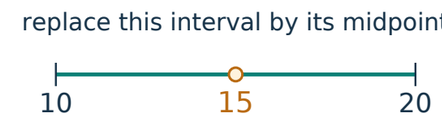
mean = (6+10)/(3+2) = 3.2

Mean=Σfx/Σf.

Remember: A value repeated f times contributes f×x to the total.

06 / 7.4

Estimate a mean from intervals



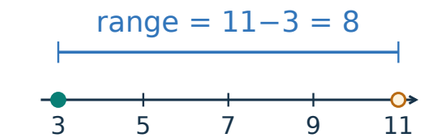
midpoint × frequency → estimated total

Midpoint=(lower+upper)/2; estimated mean=Σ(f×midpoint)/Σf.

Remember: The midpoint represents unknown individual values, so the mean is an estimate.

07 / 7.5

Use the range to find an extreme



minimum → entire spread → maximum

Range=max−min; max=min+range; min=max−range.

Remember: Check the resulting minimum and maximum against every other value.

08 / 7.5

Check whether all statistics can hold



Every condition must fit the SAME data set.

Test total, ordering, median, range and integer conditions together. One contradiction rules the claim out.

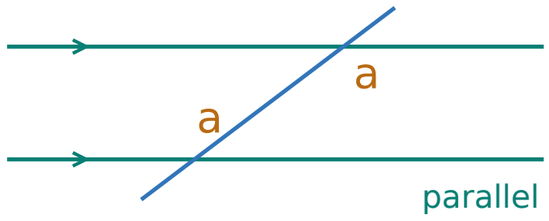
Remember: A plausible average alone does not prove a data set exists.

Geometry 1 - Triangles, Quadrilaterals and Theorems

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 8.1

Find angles on lines and transversals

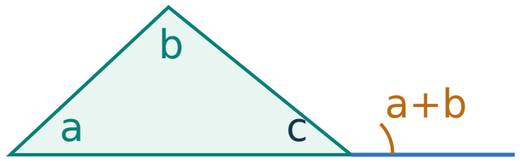


Straight=180°; turn=360°; vertical opposites equal. Parallel lines: corresponding/alternate equal; same-side interiors sum 180°.

Remember: State the parallel-line condition before using F, Z or interior-angle rules.

02 / 8.2

Link triangle interiors and an exterior angle

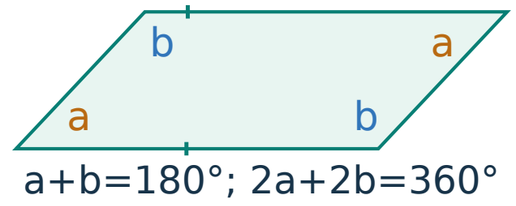


a+b+c=180°; exterior angle=a+b for the two remote interiors.

Remember: The adjacent interior is not part of the remote-angle sum.

03 / 8.3

Use only the quadrilateral properties given

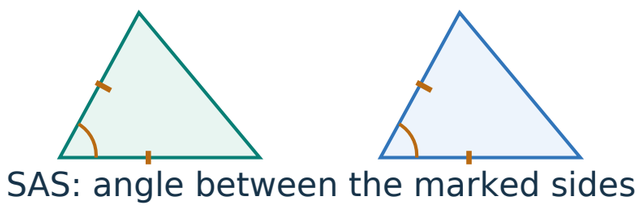


Simple quadrilateral: interior sum=360°. Parallelogram: opposite sides/angles equal; adjacent angles sum 180°.

Remember: A rhombus need not have right angles; do not add unstated properties.

04 / 8.4

Match triangles with a valid congruence test

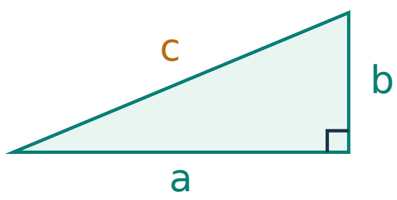


SSS; SAS (included angle); ASA; RHS (right triangles).

Remember: AAA gives similarity; SSA is not a general congruence test.

05 / 8.5

Find a side in a right triangle

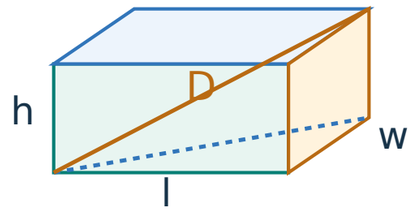


$c^2=a^2+b^2$; $c=\sqrt{(a^2+b^2)}$; $a=\sqrt{(c^2-b^2)}$.

Remember: c is opposite the 90° angle. Take the square root for a length.

06 / 8.5

Find a cuboid’s space diagonal



$d=\sqrt{(l^2+w^2)}$; $D=\sqrt{(d^2+h^2)}=\sqrt{(l^2+w^2+h^2)}$.

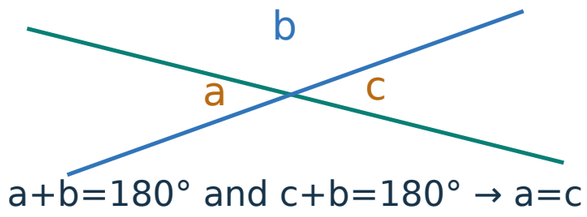
Remember: Use Pythagoras twice; an edge route is longer than the straight diagonal.

Geometry 1 - Triangles, Quadrilaterals and Theorems

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 8.6

Build a proof from accepted reasons

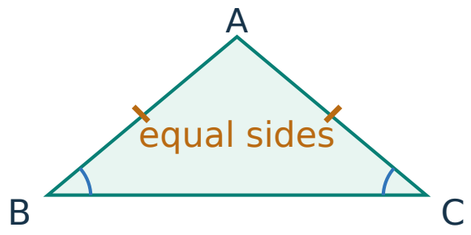


Definition=meaning; axiom=starting fact; theorem=proved fact; converse=reversed implication; corollary=direct consequence.

Remember: Equal-looking angles in a sketch do not prove equality.

08 / 8.6

Connect equal sides and opposite angles

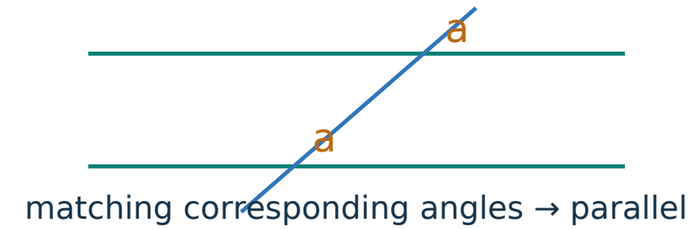


AB=AC ⇔ ∠ABC=∠ACB in triangle ABC.

Remember: Forward theorem and converse require their own justification.

09 / 8.6

Prove parallelism from matching angle positions

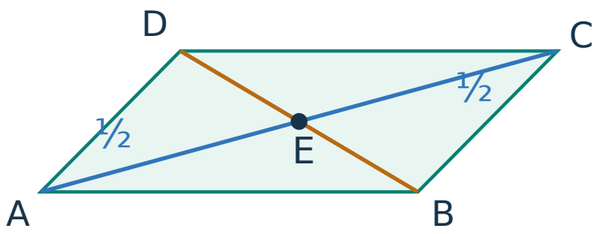


Equal alternate angles ⇒ parallel lines.
Parallel lines ⇔ equal corresponding angles.

Remember: The equal angles must occupy the required transversal positions.

10 / 8.6

Use the bisecting diagonals of a parallelogram

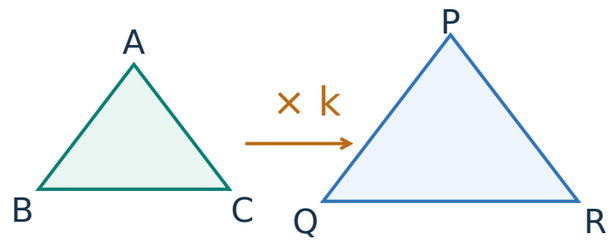


If diagonals AC and BD meet at E: AE=CE and BE=DE.

Remember: The two complete diagonals need not have equal lengths.

11 / 8.6

Match corresponding sides in similar triangles

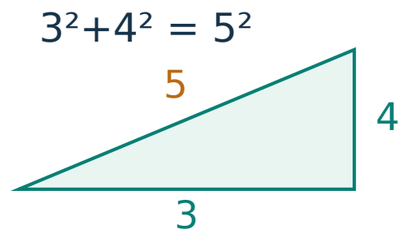


AB/PQ=AC/PR=BC/QR; A↔P, B↔Q, C↔R.

Remember: Match corresponding vertices or opposite angles before forming ratios.

12 / 8.6

Test whether a triangle is right-angled



If c is longest and $c^2=a^2+b^2$, the angle opposite c is 90°.
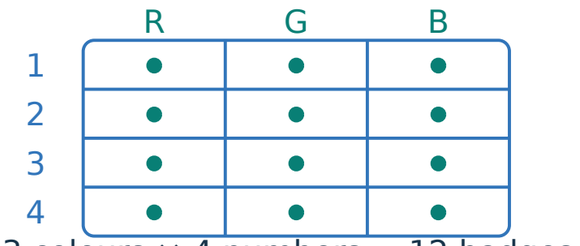
Remember: Put the longest side alone; this is the converse of Pythagoras.

Probability

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 9.1

Count all possible combinations



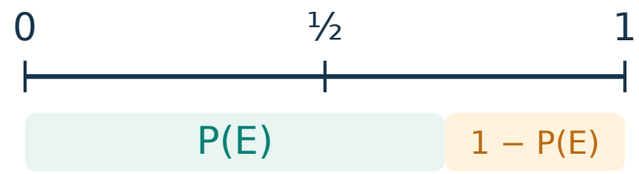
3 colours × 4 numbers = 12 badges

Total = $n_1 \times n_2 \times \ldots$

Remember: Multiply only when each earlier choice has the stated number of later choices.

02 / 9.2

Find the probability of “not”



Impossible ←→ Certain

0 ≤ P(E) ≤ 1; P(not E) = 1 − P(E).

Remember: “Not both” includes exactly one and neither.

03 / 9.3

Count favourable outcomes



4 favourable / 12 equally likely

P(E) = favourable outcomes ÷ total outcomes

Remember: Use this count only when the individual outcomes are equally likely.

04 / 9.3

Update a mixture after adding items

18 red + 30 blue

48 existing | g added

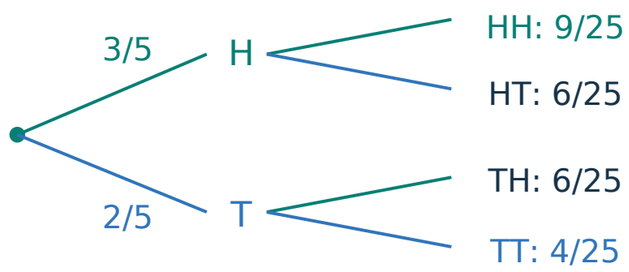
New total: 48 + g

A:B = a:b ⇒ A=ak, B=bk; new total = old total + g.

Remember: An added share is measured against the final total.

05 / 9.4

Multiply along independent paths



P(A and B)=P(A)P(B); add mutually exclusive path products.

Remember: Use unchanged branch probabilities only for independent events.

Probability

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

06 / 9.3

Count repeated labels correctly



Three “2” faces out of six

P(2) = 3/6 = 1/2

Probability follows physical faces or sector angles.

Remember: Distinct labels need not have equal probabilities.

07 / 9.5

Estimate a future count



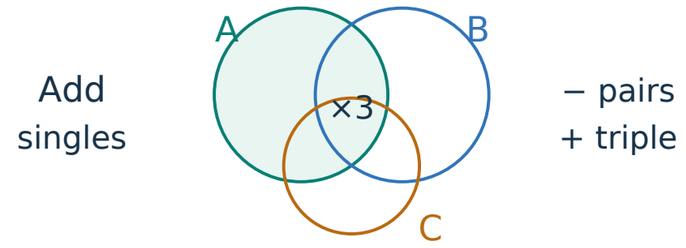
Estimate → expected count 90

Estimated P(E)=successes/trials; expected count=n × P(E).

Remember: An expectation assumes comparable conditions; it is not a guarantee.

08 / 9.6

Count overlapping sets once

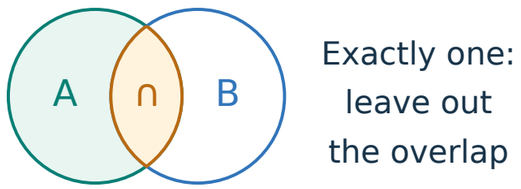


$$n(A\cup B)=n(A)+n(B)-n(A\cap B)$$
$$n(A\cup B\cup C)=n(A)+n(B)+n(C)-n(A\cap B)-n(A\cap C)-n(B\cap C)+n(A\cap B\cap C)$$

Remember: Pair intersections include the triple overlap.

09 / 9.6

Turn words into set regions



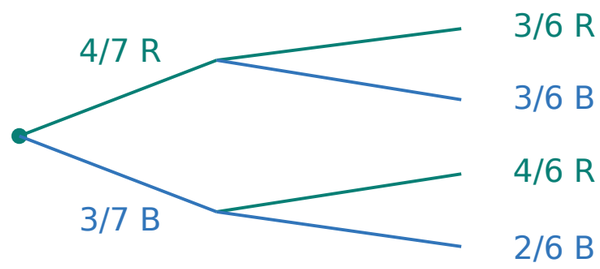
AND: overlap OR: either circle

Or = union; and = intersection; exactly one = single-set-only regions.

Remember: “At least two” includes the triple region in a three-set diagram.

10 / 9.7

Update a tree without replacement



P(A then B)=P(A) × P(B given A).

Remember: After removing one counter, update both counts and the total.

Simultaneous equations

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 10.1

Clear fractions from equations

$$\frac{x}{2} + \frac{y}{3} = 5$$

× 6 ↓

$$3x + 2y = 30$$

Multiply every term by the LCM of the denominators.

Remember: Denominators must be non-zero; multiply the right-hand side too.

02 / 10.1

Eliminate or substitute a variable

$$\begin{array}{r} 2x + y = 18 \\ x + y = 11 \\ \hline x = 7 \end{array}$$

subtract ↓

$$y = 11 - 7 = 4$$

Use equivalent operations; the answer must satisfy both equations.

Remember: Choose matching coefficients for elimination, or an isolated variable for substitution.

03 / 10.2

Build equations for two rates

6 hours × r (first rate) + 2 hours × s (second rate)

$$6r + 2s = \text{total pay}$$

Total = $\text{quantity}_1 \times \text{rate}_1 + \text{quantity}_2 \times \text{rate}_2$.

Remember: Define unknown rates and keep their units consistent.

04 / 10.2

Separate fixed cost and daily rate

3 daily charges

F (once) + d + d + d

$$C = F + 3d$$

C=F+nd; two different n-values identify F and d.

Remember: The fixed charge occurs once, not once per day.

05 / 10.2

Translate changed ages or fractions

x − k → x now

k years later

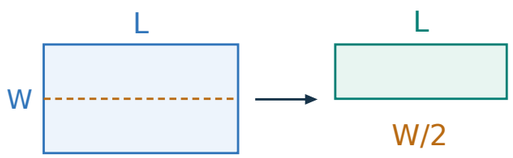
Changed fraction: (n+a)/(d+b)

k years ago: x−k; changed fraction: (n+a)/(d+b).

Remember: Require d+b≠0. Twice a past age is 2(x−k).

06 / 10.2

Find dimensions after a fold



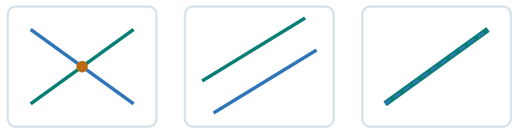
2L + 2W → 2L + W

P=2(L+W); halving W gives P_folded=2L+W.

Remember: A fold changes one dimension; it need not halve the perimeter.

07 / 10.2

Recognise how many solutions exist



One pair | No pair | Infinite

Crossing: 1 solution; distinct parallel: 0; coincident: infinitely many.

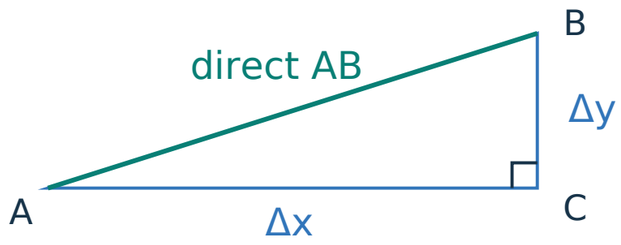
Remember: A simultaneous solution is a point on both lines.

Coordinate Geometry - The Line

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 11.1

Find a direct or multi-leg distance

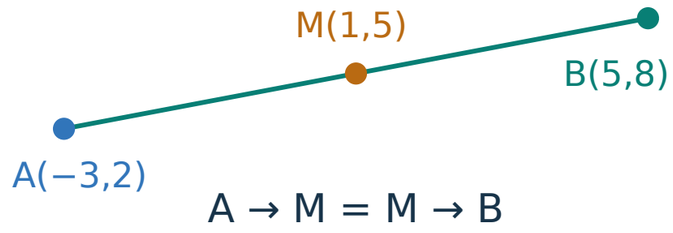


$AB=\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2]}$; via C: AC+CB.

Remember: Use the distance formula for each straight leg of a route.

02 / 11.1

Find a midpoint or missing endpoint

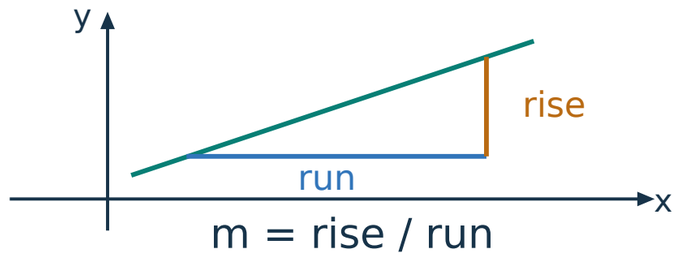


$M=((x_1+x_2)/2,(y_1+y_2)/2)$; $B=2M-A$.

Remember: Apply the missing-endpoint rule separately to each coordinate.

03 / 11.2

Find a slope and its exceptions

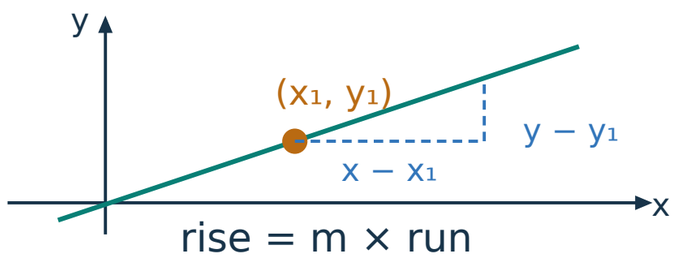


$m=(y_2-y_1)/(x_2-x_1)$; horizontal $m=0$; vertical undefined.

Remember: Require $x_2 \neq x_1$ for the fraction; keep subtraction order consistent.

04 / 11.3

Build a line from a point and slope



$y-y_1=m(x-x_1)$.

Remember: Keep coordinates grouped: subtracting a negative becomes addition.

05 / 11.4

Read slope and intercept

$$3x + 2y - 8 = 0$$

$$y = -(3/2)x + 4$$

slope $-3/2$ intercept 4

$ax+by+c=0 \Rightarrow y=-(a/b)x-c/b$; $m=-a/b$.

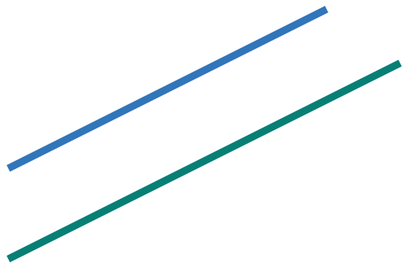
Remember: Require $b \neq 0$. Isolate y before reading off coefficients.

Coordinate Geometry - The Line

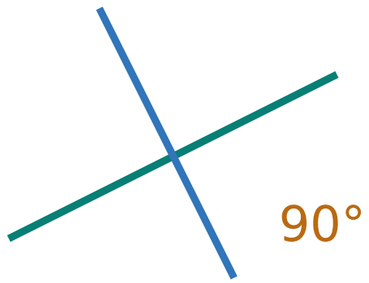
FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

06 / 11.5

Choose parallel or perpendicular slope



$m_1 = m_2$



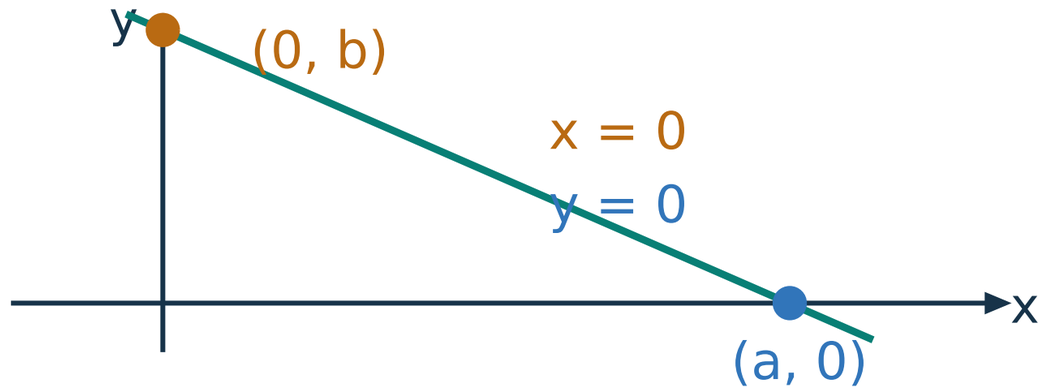
$m_1 m_2 = -1$

Parallel: $m_1=m_2$; perpendicular finite slopes: $m_1m_2=-1$. Horizontal $y=k$ has $m=0$; vertical $x=k$ is undefined.

Remember: A horizontal-vertical pair is perpendicular; do not multiply undefined slopes.

07 / 11.6

Plot a line using intercepts

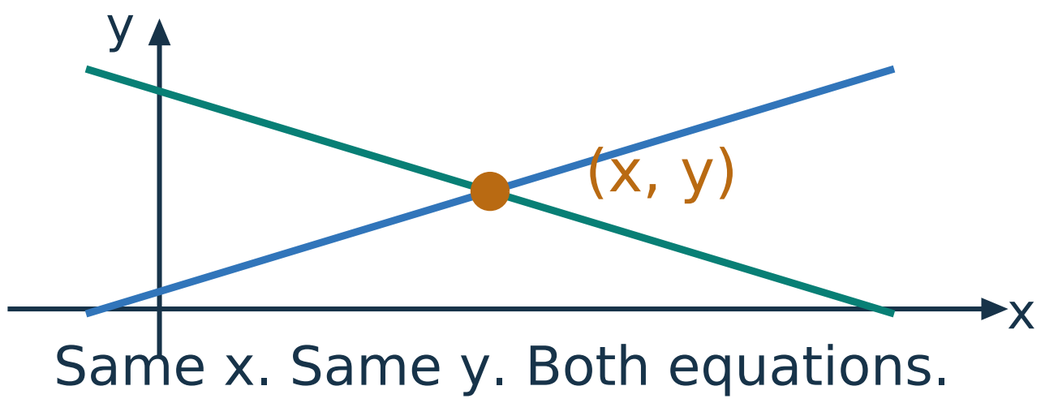


x-intercept: set y=0; y-intercept: set x=0.

Remember: If both give the origin, choose a different second point.

08 / 11.7

Find the intersection of two lines

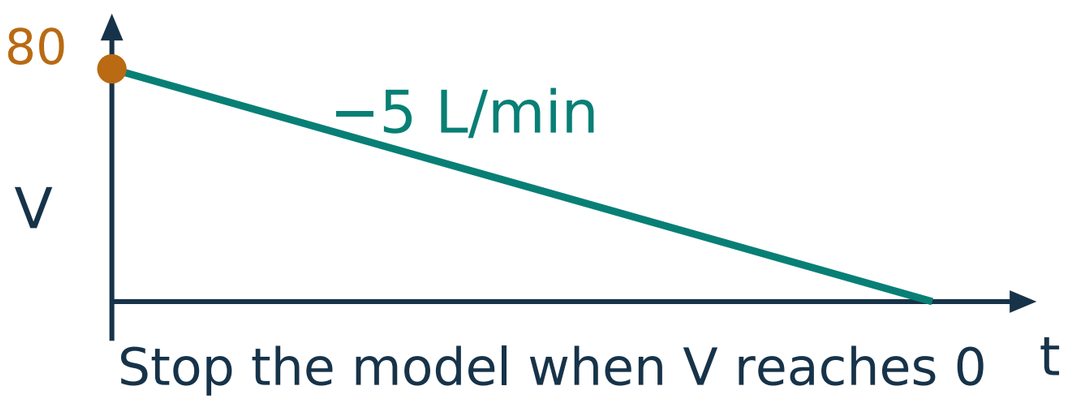


Solve the two line equations simultaneously.

Remember: Graph readings are approximate until confirmed algebraically.

09 / 11.8

Read a rate and sensible domain



y=mx+c; m=Δy/Δx; c=value at x=0.

Remember: State units and restrict the model to meaningful inputs.

Ratio - Time - Speed

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 12.1

Share a total or rebuild it from one share



Total = 8 equal parts

A:B=a:b ⇒ A=aN/(a+b), B=bN/(a+b). If A is known: one part=A/a; N=(a+b)A/a.

Remember: N is the total. For reconstruction require a≠0.

02 / 12.1

Mix batches with different ratios



Add ingredient amounts, then form a ratio

Ingredient amount=batch volume × ingredient parts/total parts.

Remember: Find ingredient amounts before combining; do not average ratios.

03 / 12.1

Distinguish direct and inverse proportion



Compare the constant relationship

Direct: y=kx. Inverse: xy=k. Equal workers × time=fixed work.

Remember: Worker-time models assume equal productivity and divisible work.

04 / 12.1

Finish a job after the team changes



66 ÷ 11 workers = 6 more days

4 elapsed + 6 more = 10 days

Remaining worker-days=total worker-days−completed worker-days.

Remember: Use the remaining work for the new team; then add elapsed time.

05 / 12.2

Measure a duration across midnight



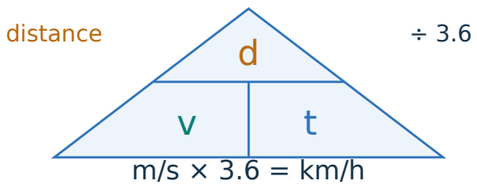
18 + 77 = 95 minutes

1 h=60 min; 1 day=24 h. Add 24 h to a next-day clock time before subtraction.

Remember: Put both times on one continuous timeline.

06 / 12.3

Choose the distance, speed or time formula



v=d/t; d=vt; t=d/v; 1 m/s=3.6 km/h.

Remember: Match time and distance units; 45 min=0.75 h.

07 / 12.3

Find the average speed of a journey



76 km ÷ 1.25 h = 60.8 km/h

Average speed=total distance/total elapsed time.

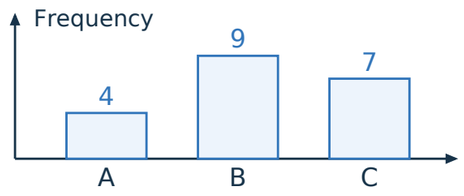
Remember: Include stops unless moving time is specifically requested.

Statistics 3 - Presenting Data

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 13.1

Read categories and frequencies



Each dot=one observation; labelled frequency-bar height=frequency.

Remember: A high bar means a frequent category, not a larger data value.

02 / 13.1

Recover a missing value from a mean

known sum: 216 | ?

10 values × mean 25 = 250

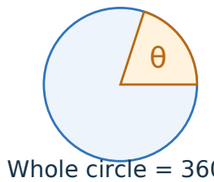
missing = 250 − 216 = 34

Mean=Σvalues/n; total=n×mean; missing=n×mean−known sum.

Remember: Use the final number of values, including the missing one.

03 / 13.2

Convert a pie angle to a share or total



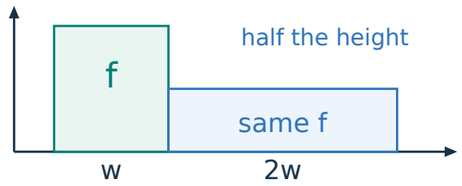
$$\frac{\text{part } f}{\text{whole } N} = \theta / 360°$$

θ=360°f/N; f=(θ/360°)N; N=360°f/θ.

Remember: Use non-zero N; recovering N also requires θ≠0.

04 / 13.3

Read frequency from a histogram



Density=f/w; frequency=bar area=density×class width.

Remember: With unequal class widths, compare areas rather than heights.

05 / 13.4

Read a stem-and-leaf display

Stem	Leaves
1	2 8 8
2	3 5 5

Key 1|2=12

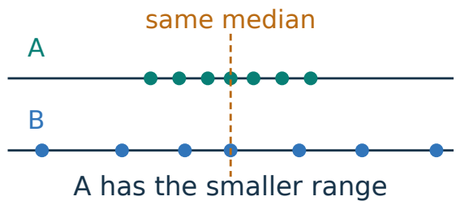
Middle values: 18 and 23

Median: position (n+1)/2 if n odd; average positions n/2 and n/2+1 if even. Mode=most frequent; range=max−min.

Remember: Read the place-value key and put leaves in order.

06 / 13.4

Compare centre and spread

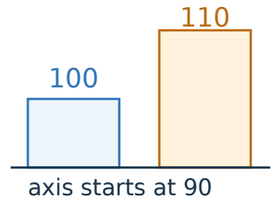


Median compares centres; range=max−min compares overall spread.

Remember: A smaller range indicates greater consistency by that measure only.

07 / 13.5

Check whether a chart misleads



Actual rise

10%

not 100%

Percentage change=(new−old)/old ×100%. A k-by-k pictogram has area factor k².

Remember: Require old≠0. Check baselines, scales and pictogram areas.

Indices - Standard form - Surds

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 14.1

Choose the correct index law

$$x^3 \times x^2$$

x × x × x | x × x

5 factors of x → x^5

$a^m a^n = a^{m+n}$; $a^m/a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $(ab)^n = a^n b^n$; $(a/b)^n = a^n/b^n$.

Remember: Exclude zero in divisors/negative powers. Use positive bases for rational-index laws.

02 / 14.2

Interpret zero, negative and fractional powers

a^0 → 1

a^{-n} → $1/a^n$

a^(m/n) → root → power

Denominator: root. Numerator: power.

$a^0=1$; $a^{-n}=1/a^n$; a^(m/n)=$(\sqrt[n]{a})^m$.

Remember: For zero/negative powers a≠0. For the stated fractional rule, a>0 and n is a positive integer.

03 / 14.3

Solve an equation with equal bases

3^(2x−1) = 27^(x−2)

3^(2x−1) = 3^[3(x−2)]

2x − 1 = 3(x − 2)

Same base → match exponents

For b>0, b≠1: $b^u=b^v \Rightarrow u=v$.

Remember: Changing 27 to 3^3 multiplies the whole outer exponent by 3.

04 / 14.4

Simplify and combine matching surds

√72 → √36 × √2

$6\sqrt{2}$

Take out the largest square factor

$\sqrt{(ab)}=\sqrt{a}\sqrt{b}$; $p\sqrt{q}+r\sqrt{q}=(p+r)\sqrt{q}$.

Remember: Use a,b≥0. Combine only matching simplified square-root parts.

05 / 14.4

Multiply surds or clear a root denominator

$$\frac{3}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

Multiply numerator AND denominator

$\sqrt{a}\sqrt{b}=\sqrt{(ab)}$; $\sqrt{a}/\sqrt{b}=\sqrt{(a/b)}$; $p/\sqrt{q}=p\sqrt{q}/q$; $(\sqrt{a}\pm\sqrt{b})^2=a+b\pm2\sqrt{(ab)}$.

Remember: a,b≥0 for products; b>0 for quotients; q>0 for rationalising.

Indices - Standard form - Surds

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

06 / 14.5

Multiply or divide standard-form numbers

$(6.4 \div 2.5) \times 10^{(9-6)}$

↓

2.56×10^{3}

$1 \leq |\text{coefficient}| < 10$

$\text{Product} = ab \times 10^{m+n}$; $\text{quotient} = (a/b) \times 10^{m-n}$, $b \neq 0$.

Remember: Normalise a non-zero answer so $1 \leq |\text{coefficient}| < 10$.

07 / 14.5

Add or subtract standard-form numbers

$$6.4 \times 10^{9} - 7.5 \times 10^{8}$$

$$6.4 \times 10^{9} - 0.75 \times 10^{9}$$

$$(6.4 - 0.75) \times 10^{9}$$

Match powers before adding coefficients

Write a common power of ten; then add or subtract coefficients.

Watch for: Do not add or subtract exponents for addition or subtraction.

08 / 14.6

Round to the requested accuracy

0.00 6 2 84 →

leading zeros keep 2 digits 8 rounds up

$0.006284 \rightarrow 0.0063$ (2 s.f.)

Decimal places count from the decimal point; significant figures start at the first non-zero digit.

Remember: Use the next digit to decide rounding; retain place value.

09 / 14.7

Enter brackets and test a claim

$(3 + 4)^2 = 49 \neq 3^2 + 4^2 = 25$

One counterexample disproves “always”

Bracket compound numerators, denominators and radicands. One valid counterexample disproves a universal claim.

Remember: One agreeing example cannot prove an identity for all values.

Quadratic equations

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 15.1

Solve by factorising

$$x^2 - x - 12 = 0$$

$$(x - 4)(x + 3) = 0$$

$x = 4$ $\quad$ $x = -3$

$uv=0 \Rightarrow u=0$ or $v=0$.

Remember: First rearrange the equation so the product equals zero.

02 / 15.2

Use the quadratic formula

$$3x^2 - 5x - 2 = 0$$

$a = 3$ $\quad$ $b = -5$ $\quad$ $c = -2$

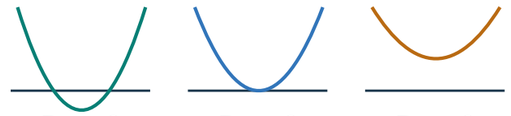
Keep the sign with each coefficient

$x=[-b\pm\sqrt{(b^2-4ac)}]/(2a)$, $a\neq0$.

Remember: Write $ax^2+bx+c=0$ first; keep the signs of a, b and c.

03 / 15.2

Predict how many real roots exist

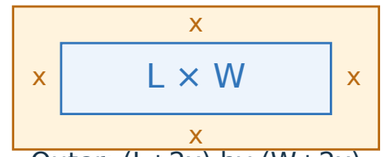


$D=b^2-4ac$: $D>0$ → two; $D=0$ → repeated; $D<0$ → no real roots.

Remember: A negative D means no real roots, not two negative roots.

04 / 15.3

Find a uniform border width



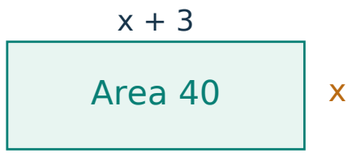
Outer: $(L+2x)$ by $(W+2x)$

Outer area=$(L+2x)(W+2x)$; border area=outer area−LW.

Remember: A border on both sides adds 2x to each dimension.

05 / 15.3

Build and check a quadratic model



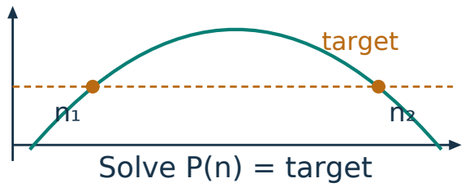
$x(x+3)=40 \rightarrow x>0$

Translate the original relation → solve → apply the domain.

Remember: Reject a root only when it violates a stated or physical restriction.

06 / 15.3

Find both times a target is reached



Set model expression=target, then rearrange to zero and solve.

Remember: Both roots may be valid; test them against the model's domain.

07 / 15.4

Build an equation from its roots

root −3 $\quad$ root 4

$$a(x+3)(x-4)=0$$

$a \neq 0$

Roots r,s: $a(x-r)(x-s)=0$; expand to $ax^2-a(r+s)x+ars=0$, $a\neq0$.

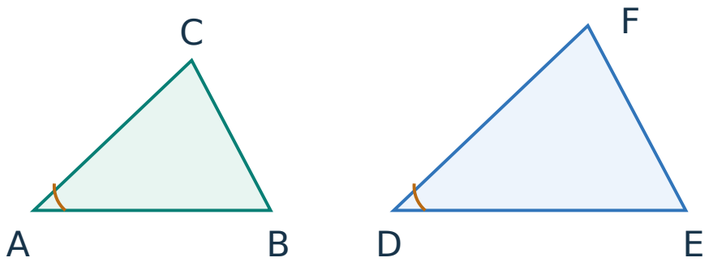
Remember: A negative root creates a plus sign in its factor.

Geometry 2 - Similar Triangles, Circles and Theorems

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 16.1

Match sides of similar triangles

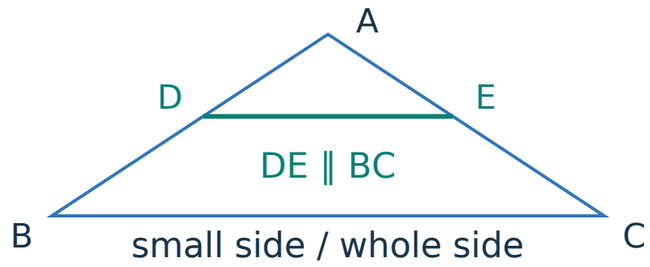


AA: two equal angle pairs suffice. $\Delta ABC \sim \Delta DEF \Rightarrow AB/DE=BC/EF=AC/DF$.

Remember: Corresponding angles are equal; all matching lengths use one scale.

02 / 16.2

Use a parallel cut in a triangle

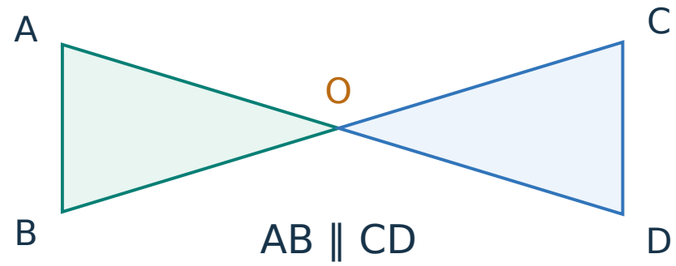


$DE \parallel BC \Rightarrow AD/AB=AE/AC=DE/BC$; $AD/DB=AE/EC$.

Remember: Match whole with whole or remaining part with remaining part; denominators $\neq 0$.

03 / 16.2

Match crossing similar triangles

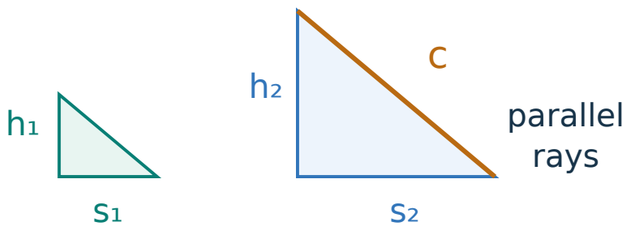


Vertically opposite angles + a parallel-line angle pair ⇒ AA similarity.

Remember: Match vertices before taking ratios, even when a triangle is reversed.

04 / 16.1

Use shadows and a right triangle

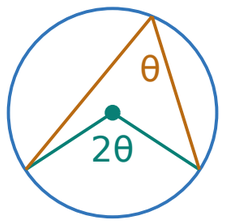


$h_1/s_1=h_2/s_2$; $c^2=a^2+b^2$; $c=\sqrt{(a^2+b^2)}$; $a=\sqrt{(c^2-b^2)}$.

Remember: Shadow ratios assume parallel sun rays on level ground; c is the hypotenuse.

05 / 16.3

Choose the correct circle-angle fact



Same arc: 2θ

Diameter: 90°

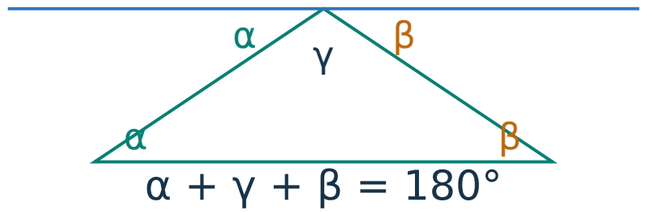
Cyclic: α+β=180°

Same arc: centre=2×edge. Same segment: edge angles equal. Diameter: edge=90°. Cyclic opposite angles sum to 180°.

Remember: Identify the same arc; the relevant centre angle may be reflex.

06 / 16.4

Prove the triangle-angle sum



α+β+γ=180° in a Euclidean triangle.

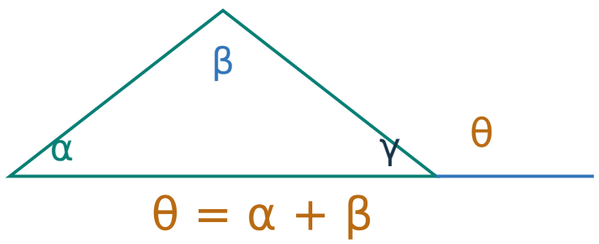
Remember: Use an auxiliary parallel line and known angle facts, not the theorem being proved.

Geometry 2 - Similar Triangles, Circles and Theorems

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 16.4

Prove the exterior-angle theorem

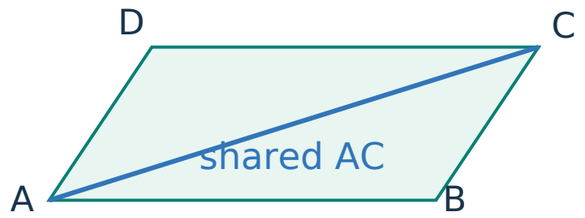


Exterior angle θ=α+β, the two non-adjacent interior angles.

Remember: Do not include the adjacent interior angle in the remote-angle sum.

08 / 16.4

Prove opposite parallelogram parts equal



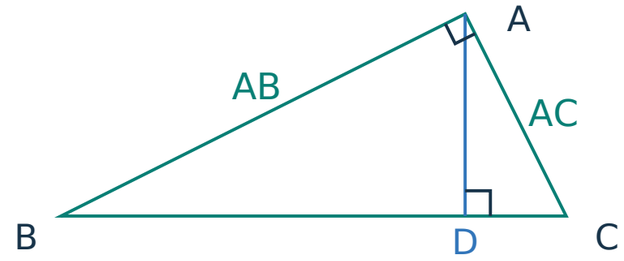
parallel angles + common side → ASA

AB∥CD and BC∥AD; diagonal AC gives ΔABC≅ΔCDA by ASA.

Remember: A parallelogram diagonal need not bisect each corner angle.

09 / 16.4

Prove Pythagoras with an altitude

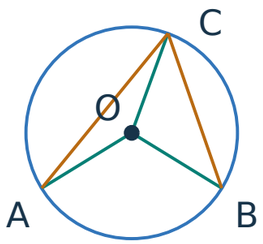


$AB^2+AC^2=BC^2$; BC is the hypotenuse.

Remember: Drop AD perpendicular to BC; D need not be its midpoint.

10 / 16.4

Prove the centre-angle theorem



OA = OB = OC

equal base angles

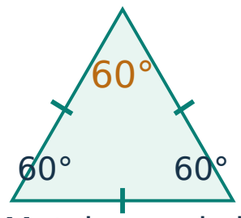
centre = 2 × edge

Central angle on the same arc = 2∠ACB.

Remember: Keep the chosen arc fixed; some cases use a reflex central angle.

11 / 16.1

Use equal sides and angle structure





Match equal sides to opposite angles

Isosceles: equal sides ⇒ equal opposite angles. Equilateral: all sides equal, angles 60°. Parallelogram: opposite parts equal, adjacent angles sum to 180°.

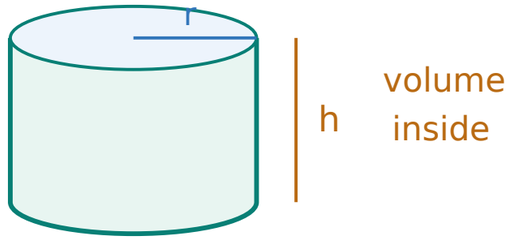
Remember: Equal adjacent parallelogram sides alone do not make equilateral triangles.

Cylinder - Sphere

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 17.1

Find a cylinder's volume, height or radius



V = πr²h; h = V/(πr²); r = √[V/(πh)], with r,h > 0.

Watch for: Using diameter as radius, or taking a square root before isolating r².

02 / 17.1

Choose the cylinder surfaces

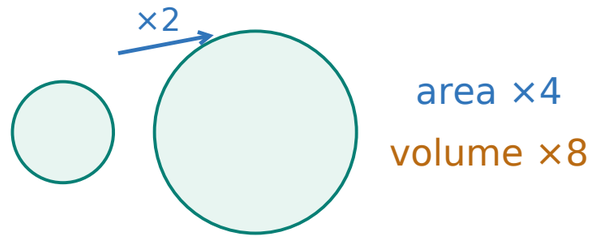


Curved A = 2πrh; open top with base: A = 2πrh + πr²; closed: A = 2πrh + 2πr².

Watch for: Adding a missing lid or confusing surface area with capacity.

03 / 17.1

Predict the effect of scaling



Cylinder V ∝ r²h; sphere V ∝ r³. With h fixed, r × k gives V × k².

Remember: Cylinder height fixed: radius ×k means volume ×k². Sphere: volume ×k³.

04 / 17.2

Sphere or hemisphere: area or volume?

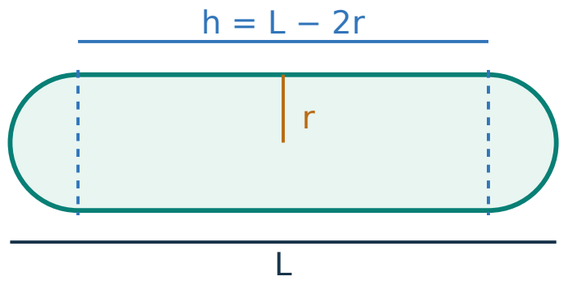


Sphere: V = 4πr³/3, A = 4πr². Hemisphere: V = 2πr³/3; curved A = 2πr²; closed A = 3πr².

Watch for: Adding a base at a hemisphere-cylinder join.

05 / 17.2

Split a capsule into familiar solids



For total end-to-end length L and radius r: h = L − 2r; V = πr²h + 4πr³/3; external A = 2πrh + 4πr², L > 2r.

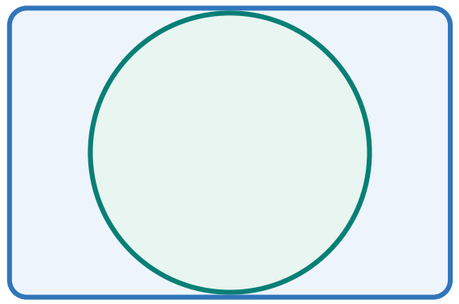
Watch for: Using L as the cylinder height.

Cylinder - Sphere

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

06 / 17.2

Find empty space or complete litres



container

− solid

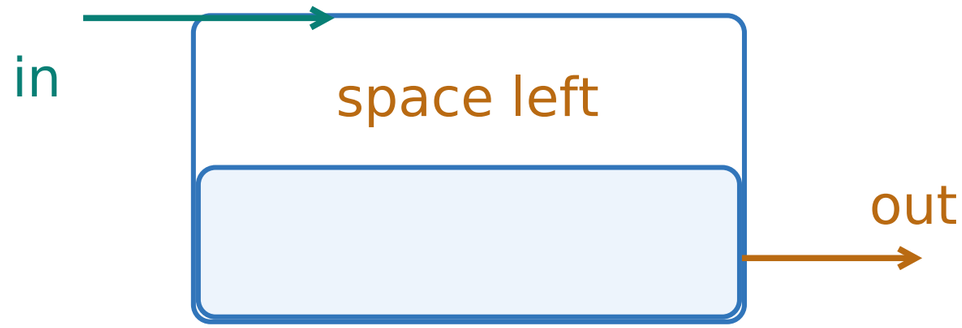
= space

Empty volume = container volume − enclosed solid volume; 1 L = 1000 cm³; complete litres = floor(capacity in L).

Watch for: Rounding 3.9 litres up to four complete litres.

07 / 17.3

Find the time left to fill a tank

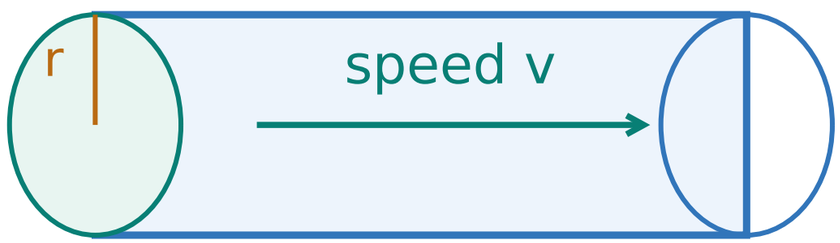


t = remaining volume / net inflow; net inflow = inflow − outflow; use compatible units.

Watch for: Using full capacity when the vessel is partly full.

08 / 17.3

Turn pipe speed into volume flow



swept volume each second

Q = cross-sectional area × speed = πr²v. For cm and s, Q is cm³/s.

Watch for: Treating a speed in cm/s as though it were a flow in litres/s.

09 / 17.3

Convert volume and time units



1 L = 1000 cm³ = 1000 mL

1 m³=10⁶ cm³=1000 L; 1 L=1000 cm³; 1 mL=1 cm³; 1 min=60 s. Area conversion squares a length factor; volume conversion cubes it.

Watch for: Using 1 m³=100 cm³ by applying only the linear conversion factor.

Patterns and Sequences

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 18.1

Separate position from term value



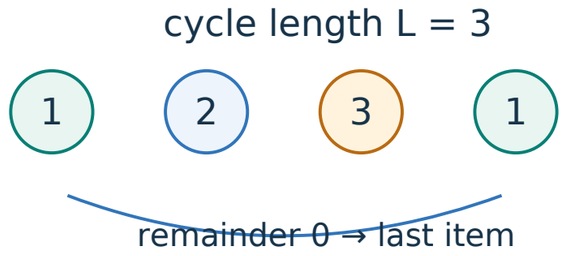
Input a position, not the previous term.

T_n is the value at position n; n is not generally the term value.

Watch for: Feeding a previous term into an nth-term rule.

02 / 18.2

Find a position in a repeating cycle

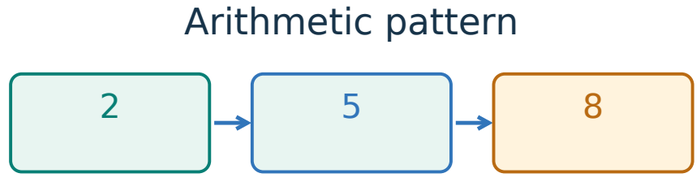


n = qL + r, 0 ≤ r < L; use cycle item r, except r = 0 means item L.

Watch for: Using zero remainder as the first item.

03 / 18.3

Recognise arithmetic or geometric growth

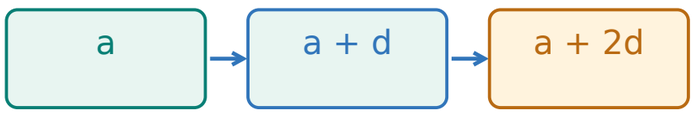


Arithmetic: $T_{n+1} - T_n = d$. Geometric: $T_{n+1}/T_n = r$ where the denominator is non-zero.

Watch for: Calling every increasing pattern linear.

04 / 18.4

Build or reverse an arithmetic rule

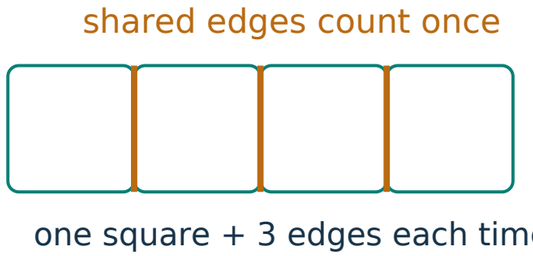


$T_n = a + (n-1)d$; for $d \neq 0$, $n = 1 + (T-a)/d$.

Watch for: Accepting n = 14.2 as a position in an integer-indexed sequence.

05 / 18.5

Count a growing shape efficiently



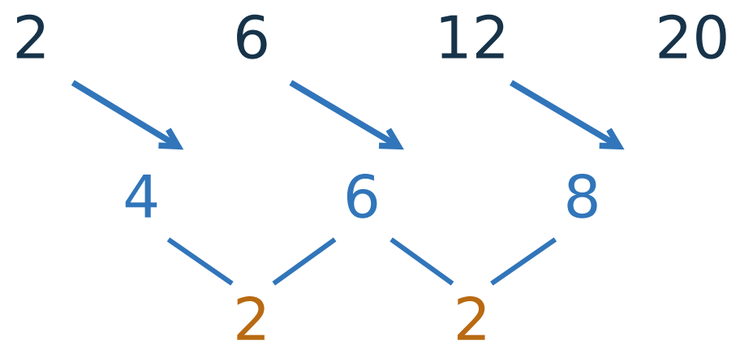
Total = fixed part + repeated part; count a shared edge once.

Watch for: Using the number of sides per shape without allowing for shared sides.

Patterns and Sequences

06 / 18.6

Find a quadratic nth-term rule



$T_n = an^2+bn+c$; constant second difference = 2a for unit-spaced n.

Watch for: Using the second difference itself as a.

07 / 18.6

Find the position of a quadratic term



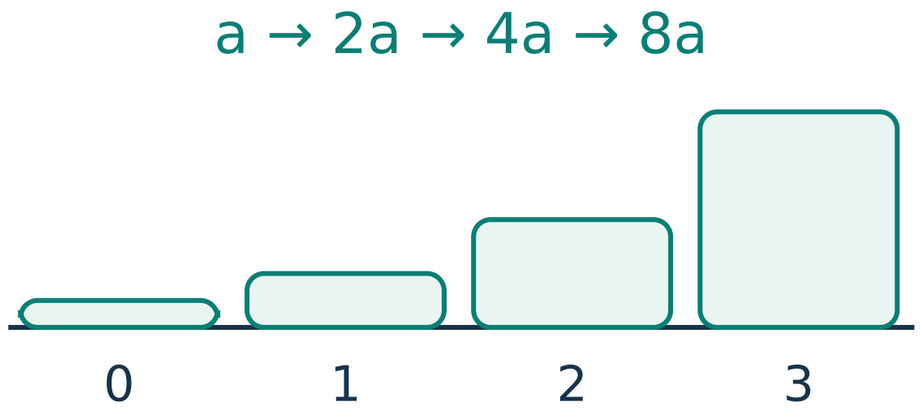
Check each candidate in the original rule.

Solve an^2+bn+c = target, then retain allowed integer n. Three independent term conditions determine a general quadratic.

Watch for: Assuming two starting values uniquely determine a quadratic sequence.

08 / 18.7

Choose the exponent from the starting stage



If stage 1 is a: $T_n = ar^{n-1}$. If stage 0 is a: $T_n = ar^n$. For positive a,r, $T_{n+k}/T_n = r^k$.

Watch for: Using ar^n when the supplied amount belongs to stage 1.

09 / 18.7

Plot a sequence as separate points



Plot (n,T_n) at allowed integer n; linear, quadratic and exponential growth have different difference patterns.

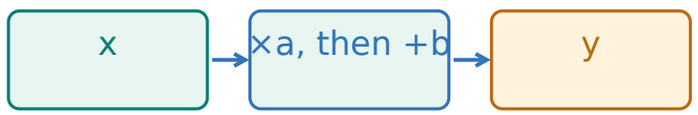
Watch for: Joining sequence points to suggest non-integer stages exist.

Functions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 19.1

Reverse a function machine



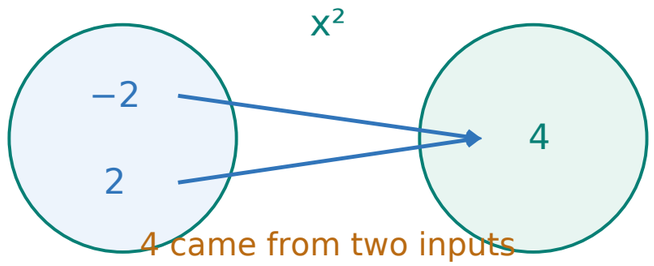
Return: subtract b, then divide by a.

For f(x)=ax+b with a ≠ 0: $f^{-1}(x)=(x-b)/a$.

Watch for: Using the original operation order when reversing.

02 / 19.1

Check whether an inverse is a function

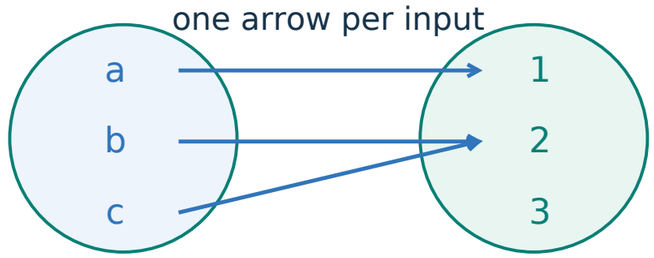


An inverse on the range is a function only when each reached output has a unique original input.

Watch for: Writing ±√x as though it were a single-valued inverse function.

03 / 19.2

Identify domain, range and function validity

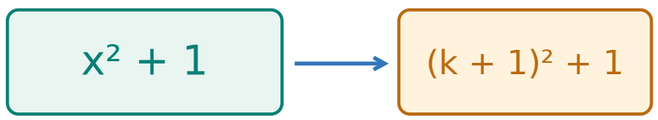


Every x in the domain has exactly one output; range = set of reached outputs ⊆ codomain.

Watch for: Rejecting many-to-one mappings.

04 / 19.3

Substitute an entire input



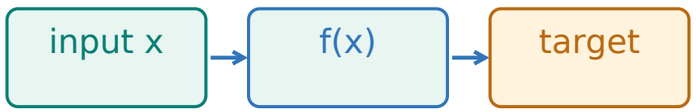
replace every x by the whole input

Replace every x by the whole input u: f(u). In general f(ku) ≠ kf(u).

Watch for: Dropping a negative sign or a cross term from a squared bracket.

05 / 19.3

Find an input for a target output



Solve for the input that reaches the target.

Replace function notation by the corresponding rules; solve in the allowed domain.

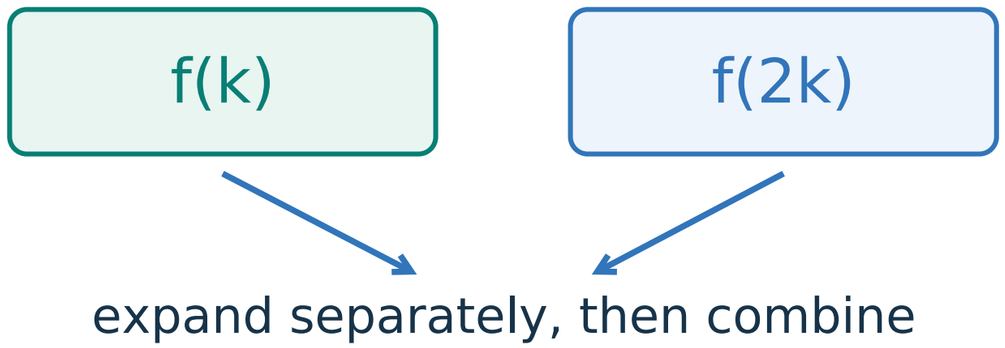
Watch for: Reporting the common output when the question asks for the input.

Functions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

06 / 19.3

Expand separate function values first

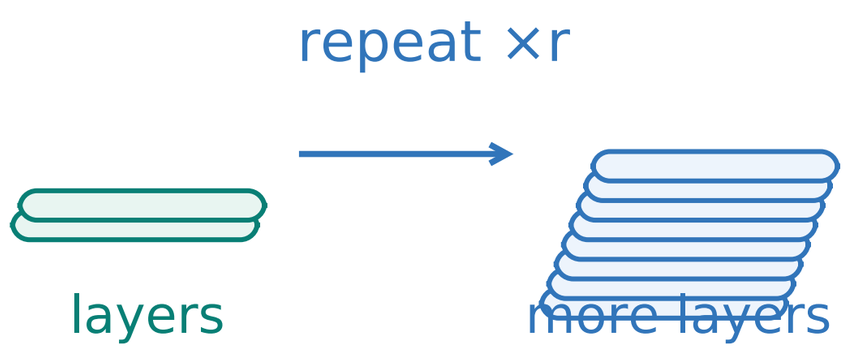


Expand each function value separately; evaluate fixed inputs before solving.

Watch for: Using f(k)+f(2k)=f(3k) without a valid reason.

07 / 19.3

Model repeated layers or thickness

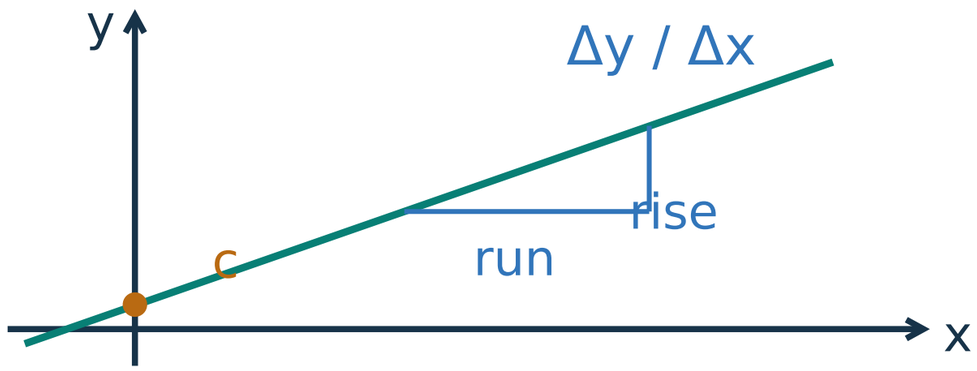


With n operations after the original sheet: $L(n)=r^n$; thickness $H(n)=h_0r^n$.

Watch for: Using the cut count as the number of layers or mixing millimetres and centimetres.

08 / 19.4

Recover a straight-line function

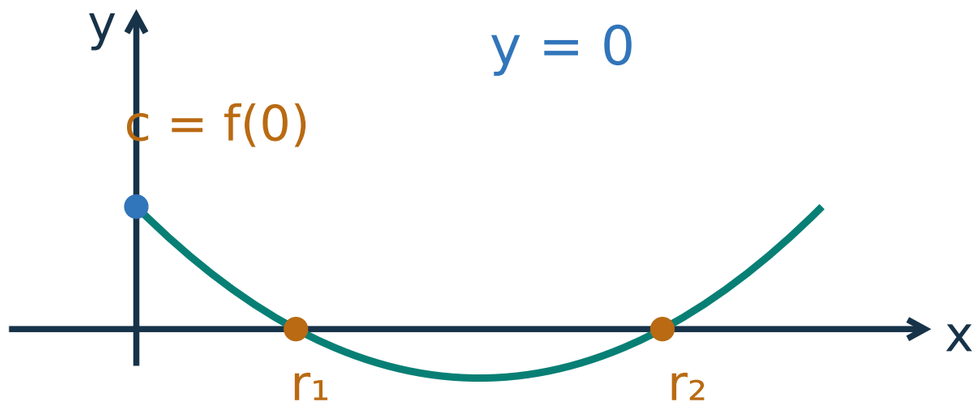


$a=(y_2-y_1)/(x_2-x_1)$, $x_2\neq x_1$; $b=y_1-ax_1$.

Watch for: Swapping input and output during substitution.

09 / 19.4

Recover a quadratic from graph clues



f(0)=c; an x-intercept solves f(x)=0. General a,b,c need three independent conditions unless a coefficient is already known.

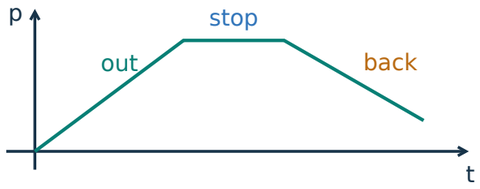
Watch for: Assuming a downward parabola cannot have positive outputs.

Drawing and Interpreting Real-life Graphs

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 20.1

Read movement, stops and return journeys

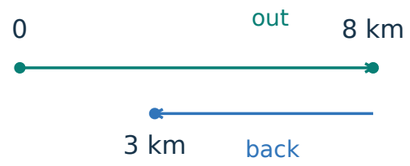


On a segment: signed rate = Δposition/Δtime; speed = |signed rate|.

Watch for: Treating a flat graph at 8 km as movement at 8 km/h.

02 / 20.1

Find total distance and average speed

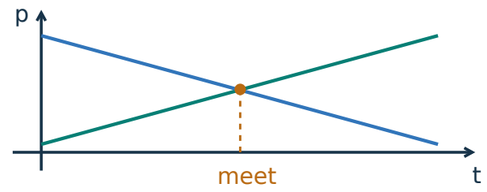


Total distance = Σ|successive position changes|; average speed = total distance/full elapsed time.

Watch for: Calling final position the total distance.

03 / 20.1

Find a meeting or separation

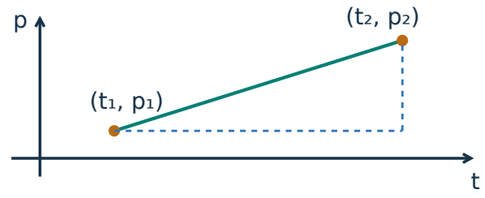


Meeting: pA(t)=pB(t); separation: |pA(t)−pB(t)| on the same line.

Watch for: Accepting an intersection outside its segment.

04 / 20.1

Write the rule for a graph segment

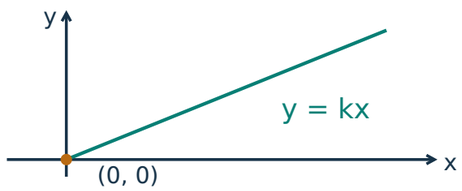


$p(t)=p_1+[(p_2-p_1)/(t_2-t_1)](t-t_1)$, for $t_1 \le t \le t_2$.

Watch for: Assuming every segment begins at the origin.

05 / 20.2

Recognise direct proportion

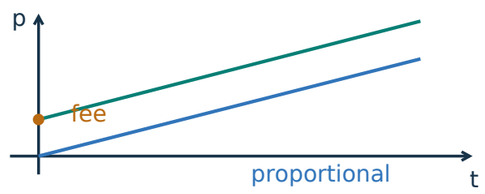


y=kx; y/x=k for x≠0; graph passes through (0,0).

Watch for: Calling y=3.5x+6 directly proportional because it is straight.

06 / 20.2

Separate a fixed fee from a rate

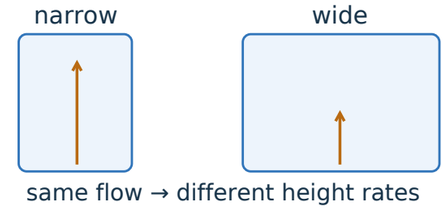


Linear: y=mx+c; directly proportional only when c=0.

Watch for: Scaling a total price while ignoring its fixed fee.

07 / 20.3

Predict how fast water height rises

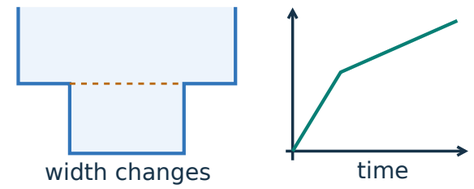


Height rate = Q/A at that level; larger cross-sectional area gives a smaller height rate.

Watch for: Assuming constant flow always gives a straight height-time graph.

08 / 20.3

Join the stages of a filling graph



Time in each part = part volume/Q; cumulative height and time give graph corners.

Watch for: Restarting time when the width changes.

Algebra 2

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 21.1

Add or subtract algebraic fractions



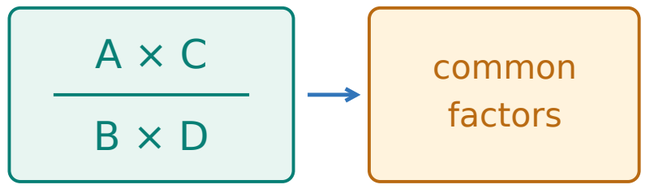
scale each entire numerator

A/B ± C/D=(AD±BC)/(BD), B,D≠0; use a smaller common denominator when possible.

Watch for: Subtracting only the first term of a bracketed numerator.

02 / 21.1

Multiply fractions by cancelling factors



(A/B)(C/D)=AC/(BD); only common non-zero factors cancel.

Watch for: Cancelling terms inside a sum.

03 / 21.1

Divide by an algebraic fraction



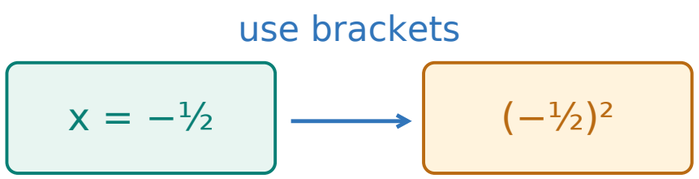
multiply by the reciprocal

(A/B)÷(C/D)=(A/B)(D/C), with B,D≠0 and C≠0.

Watch for: Checking denominator restrictions but forgetting division by zero.

04 / 21.1

Substitute a negative or fractional value



square the whole signed input

Substitute the whole value in brackets; division by a fraction multiplies by its reciprocal.

Watch for: Treating division by 3/2 as multiplication by 3/2.

05 / 21.2

Clear denominators in an equation



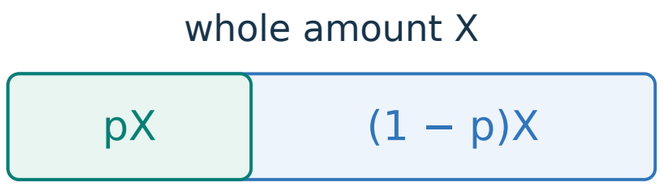
every term on both sides

Multiply both complete sides by the least common denominator; variable factors used must be non-zero.

Watch for: Multiplying only one term on a side.

06 / 21.3

Turn a fraction story into an equation



used part + remaining part

A fraction p of an amount X is pX; the remainder is (1−p)X.

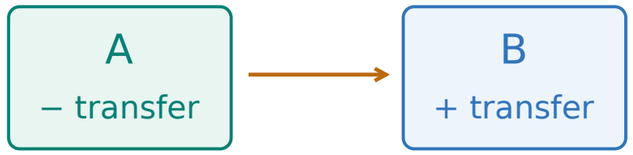
Watch for: Applying every fraction to the original amount regardless of the event order.

Algebra 2

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 21.3

Track money through successive transfers



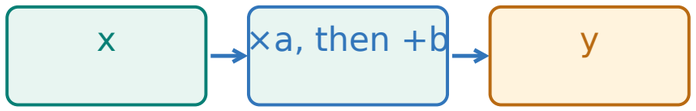
total before = total after

Update both balances after each transfer; total money is conserved if nothing enters or leaves.

Watch for: Taking B’s later transfer as a fraction of the original 30 only.

08 / 21.4

Make a variable the subject



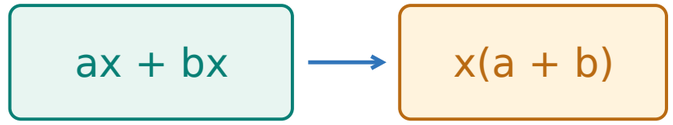
undo: −b, then ÷a

Undo operations in reverse order, keeping equality; division by a coefficient requires it to be non-zero.

Watch for: Changing signs by a slogan without applying the same operation to both sides.

09 / 21.4

Collect a subject that appears twice



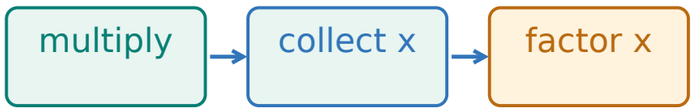
one copy of x outside the bracket

ax+bx=x(a+b); isolate by factoring, then divide only if the factor is non-zero.

Watch for: Trying to isolate two copies of x independently.

10 / 21.4

Rearrange a fractional formula



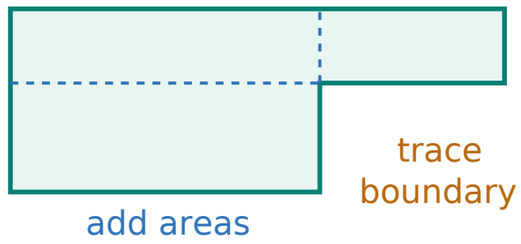
retain both denominator restrictions

If y=(ax+b)/(cx+d), then x=(b−dy)/(cy−a), provided original cx+d≠0 and cy−a≠0; exceptional cases must be checked.

Watch for: Dividing by an expression that could be zero without checking that case.

11 / 21.5

Build a formula from combined shapes

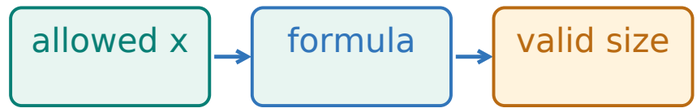


Perimeter counts the exposed boundary; area adds component areas or subtracts removed areas. All physical lengths must be positive.

Watch for: Subtracting the cut-out perimeter from the outer perimeter.

12 / 21.5

Use and reverse a formula within its domain



reject impossible physical answers

Substitution evaluates the rule; reversing solves an equation. A model’s domain may be narrower than all real numbers.

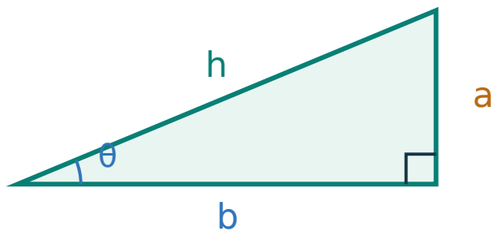
Watch for: Keeping an algebraic answer that makes a physical length non-positive.

Trigonometry

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 22.1

Find a missing side with Pythagoras

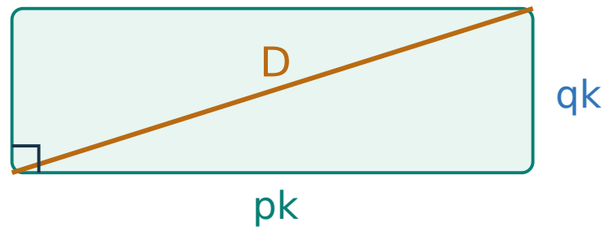


$h = \sqrt{(a^2+b^2)}$; $a = \sqrt{(h^2-b^2)}$; $h^2 = a^2+b^2$.

Remember: Right-angled triangle only. Hypotenuse h is longest; $h>b>0$ when finding a.

02 / 22.1

Use a diagonal and an aspect ratio

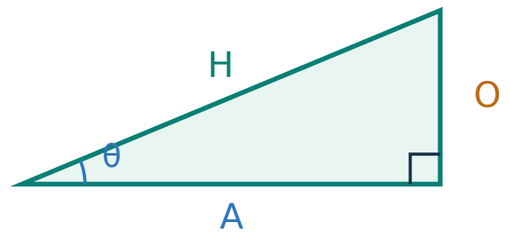


width=pk, height=qk, $k=D/\sqrt{(p^2+q^2)}$; area=$D^2pq/(p^2+q^2)$.

Watch for: Assuming the longer diagonal always means the larger area for different aspect ratios.

03 / 22.2

Label opposite, adjacent and hypotenuse

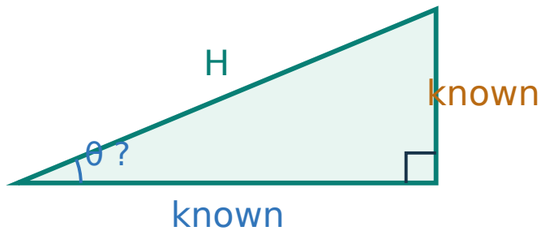


$\sin\theta=O/H$; $\cos\theta=A/H$; $\tan\theta=O/A$. O and A depend on θ; H does not.

Watch for: Calling the bottom side adjacent without looking at the angle.

04 / 22.3

Find an angle from two sides

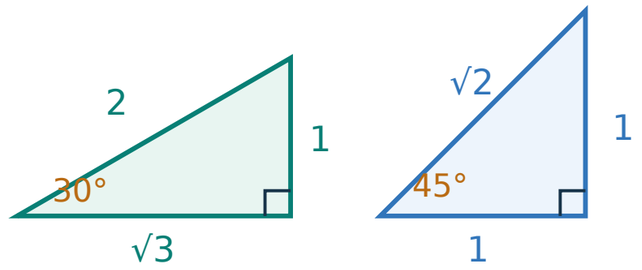


$\theta=\sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$, in degree mode.

Watch for: Reading $\sin^{-1}$ as 1/sin or using radian mode.

05 / 22.3

Keep special-angle ratios exact

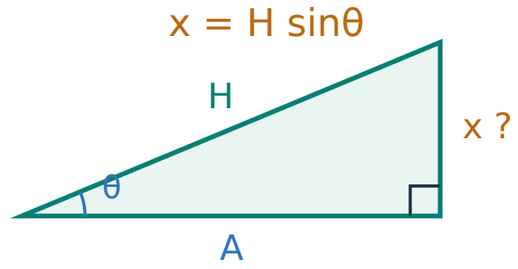


$\sin30°=1/2$, $\cos30°=\sqrt{3}/2$, $\tan30°=\sqrt{3}/3$; $\sin45°=\cos45°=\sqrt{2}/2$, $\tan45°=1$; $\sin60°=\sqrt{3}/2$, $\cos60°=1/2$, $\tan60°=\sqrt{3}$.

Watch for: Replacing √3 by 1.7 before later calculations.

06 / 22.4

Find a side above the fraction bar



If $\sin\theta=x/H$, then $x=H\sin\theta$; similarly for cosine or tangent.

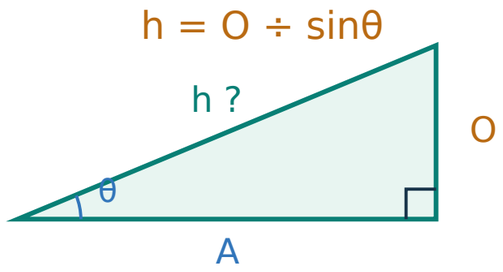
Watch for: Choosing multiplication from memory without first writing the equation.

Trigonometry

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 22.4

Find a side below the fraction bar

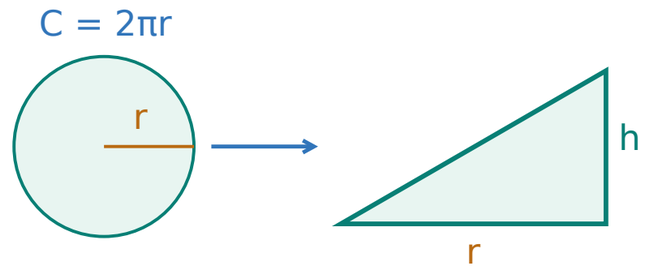


If sinθ=O/h, then h=O/sinθ; if tanθ=O/a, then a=O/tanθ.

Watch for: Multiplying 7 by sin30° when the unknown is underneath.

08 / 22.5

Convert a curved measurement to a distance

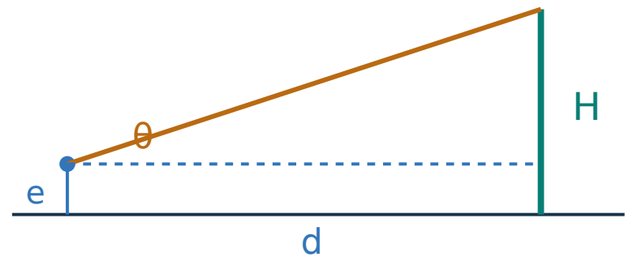


C=2πr; r=C/(2π). The right triangle needs horizontal distance, not curved boundary length.

Watch for: Using the circumference as the triangle’s horizontal leg.

09 / 22.5

Add eye height to the calculated rise

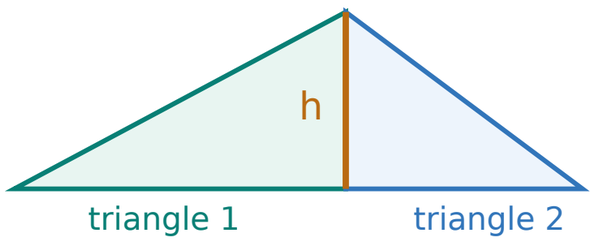


H=e+d tanθ when e is eye height and d is horizontal distance; adjust the vertical reference if ground levels differ.

Watch for: Adding eye height twice or confusing a sloping sight line with d.

10 / 22.5

Link triangles through a shared side

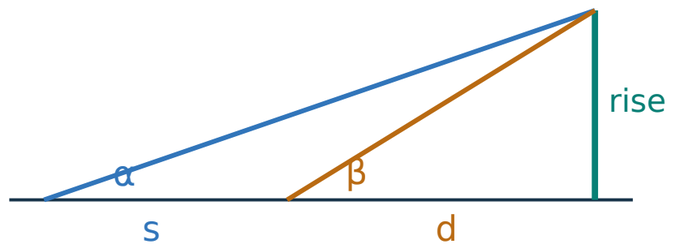


Carry the shared side exactly; (a√b)²=a²b. Each triangle needs its own right angle and reference angle.

Watch for: Adding angles without checking the rays’ order.

11 / 22.5

Link two observation positions



Same vertical rise gives d tanβ=(d+s)tanα when the farther observer is s farther away, at the same eye level.

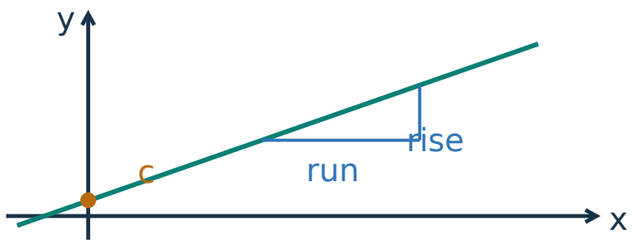
Watch for: Treating the distance between observers as either complete triangle base.

Graphing Functions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 23.1

Build a straight-line graph



m=Δy/Δx for Δx≠0; c=f(0); y−y₁=m(x−x₁).

Watch for: Plotting the constant c on the x-axis.

02 / 23.1

Recover a linear rule from a table

x	y
2	11
5	20

→

Δy = 9

Δx = 3

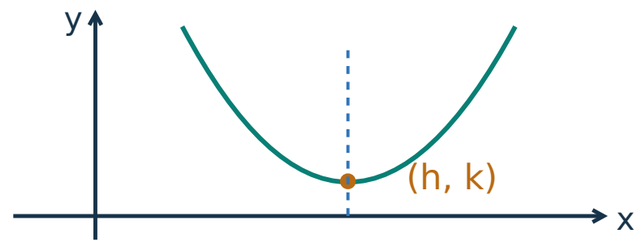
m = 3

m=(output change)/(input change); fixed part c=y−mx.

Watch for: Dividing total cost by hours when there is a fixed fee.

03 / 23.2

Find and plot a quadratic's turning point

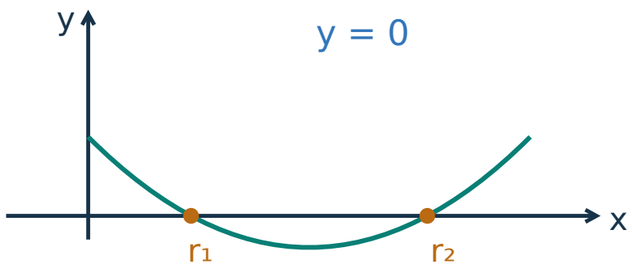


y=ax²+bx+c, a≠0; vertex x=−b/(2a), y=f(−b/(2a)); y=a(x−h)²+k has vertex (h,k).

Watch for: Drawing a quadratic as a broken straight-line graph.

04 / 23.2

Find roots and axis intercepts

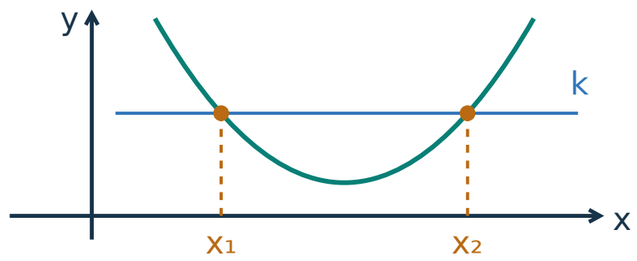


y-intercept: x=0; x-intercepts: f(x)=0. For a quadratic, factor when possible and use the zero-product rule.

Watch for: Calling a root (0,2) instead of (2,0).

05 / 23.3

Read inputs for a target output



Draw the horizontal line y=k; read all intersection x-values.

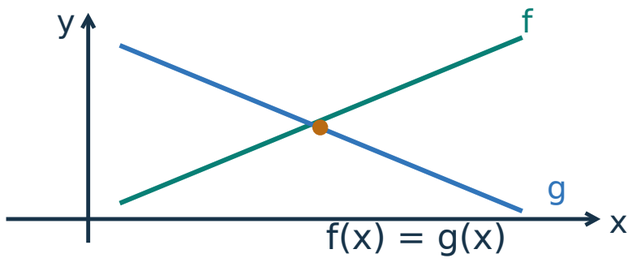
Watch for: Drawing x=k or missing the second crossing.

Graphing Functions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

06 / 23.3

Find where two functions agree

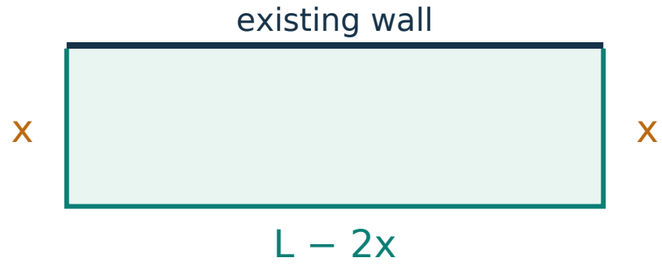


f(x)=g(x) at their intersections; equal coordinates use the same x and y.

Watch for: Confusing an intersection of two functions with an x-axis root.

07 / 23.4

Build an area model with three fenced sides

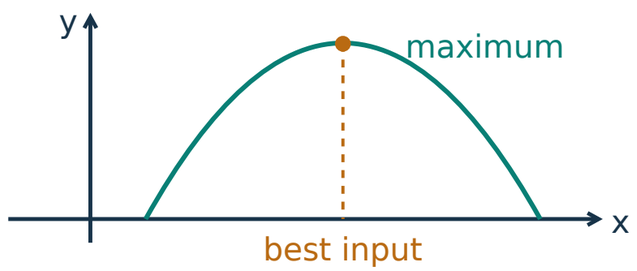


For three fenced sides with total L: length=L−2x; A=x(L−2x), with 0<x<L/2.

Watch for: Using a four-sided perimeter when one side is an existing wall.

08 / 23.4

Find and interpret a maximum or minimum

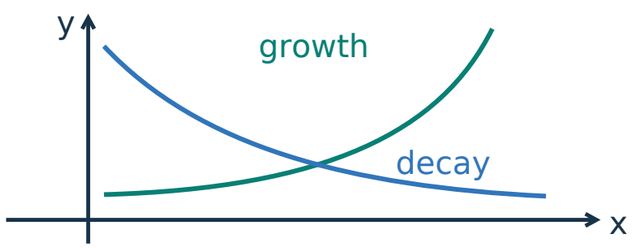


For $A=ax^2+bx+c$, vertex input=$-b/(2a)$; check domain endpoints when included. Three-side fencing: maximum $L^2/8$ at x=L/4, length=L/2.

Watch for: Giving the maximum’s input instead of the maximum output.

09 / 23.5

Recognise exponential growth or decay

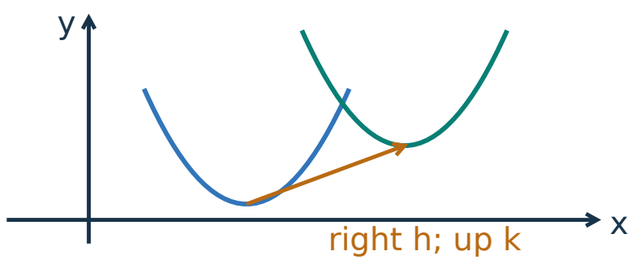


$y=ab^x$ with $a>0, b>0, b\neq1$; $y(0)=a$, $y(x+1)/y(x)=b$. $b>1$ gives growth; $0<b<1$ gives decay; $b^{-n}=1/b^n$.

Watch for: Expecting equal additive gaps or treating a graph estimate as exact.

10 / 23.5

Transform a familiar graph



f(x)+k: up k; f(x−h): right h; −f(x): reflect in x-axis; f(−x): reflect in y-axis. Combined g(x)=a f(x−h)+k maps (u,v) to (u+h,av+k).

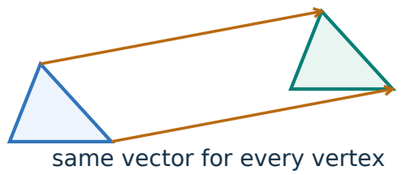
Watch for: Reading x−2 as a shift left instead of right.

Geometry 3: Transformations - Constructions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 24.1

Translate every vertex by the same vector

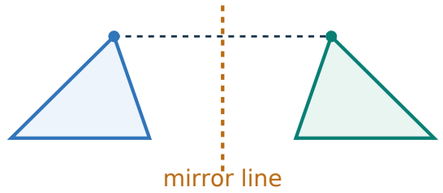


Vector (a,b): (x,y)↦(x+a,y+b); image−original=(a,b) for every vertex.

Watch for: Finding a vector from one vertex but using a different one for another.

02 / 24.2

Reflect in the stated mirror line

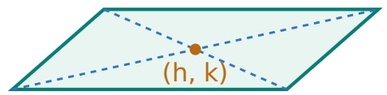


In x=a: (x,y)↦(2a−x,y); in y=b: (x,y)↦(x,2b−y); in y=x: (y,x); in y=−x: (−y,−x).

Watch for: Accepting a mirror that works for one pair but fails for the remaining vertices.

03 / 24.2

Distinguish central and axial symmetry



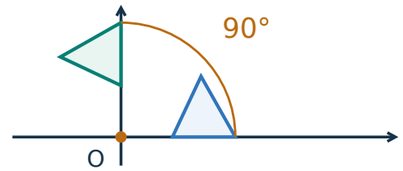
half-turn about the centre

Central symmetry about (h,k): (x,y)↦(2h−x,2k−y). Axial symmetry means reflection in a line maps the entire shape to itself.

Watch for: Treating central symmetry as reflection in an arbitrary line through the centre.

04 / 24.3

Rotate about the correct centre



About origin: 90° anticlockwise (−y,x), 90° clockwise (y,−x), 180° (−x,−y). About (h,k), translate to the origin, rotate, then translate back.

Watch for: Giving only “a rotation” or rotating around the wrong centre.

05 / 24.3

Describe a complete transformation

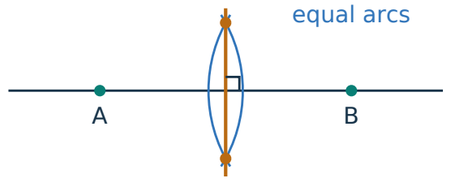


Translation needs a vector; reflection needs a mirror line; rotation needs centre, angle and direction. An ordered mapping fixes the vertex correspondence.

Watch for: Calling a point match enough evidence for a whole-shape mapping.

06 / 24.4

Construct a perpendicular bisector

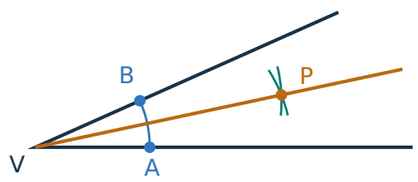


Use equal radius r>AB/2 from A and B; joining the two intersections gives the perpendicular bisector.

Watch for: Changing the compass opening between A and B.

07 / 24.4

Bisect an angle with equal arcs

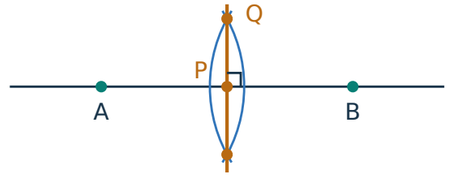


From vertex V mark VA=VB; find P with PA=PB using equal arcs; ray VP bisects ∠AVB.

Watch for: Joining A to B instead of V to P.

08 / 24.4

Construct a perpendicular on a line



Mark PA=PB on opposite sides; a point Q with QA=QB gives PQ perpendicular to AB.

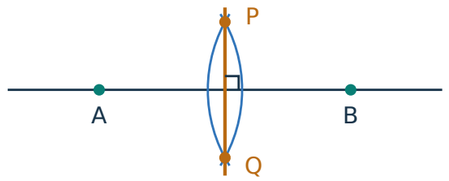
Watch for: Placing A and B at unequal distances from P.

Geometry 3: Transformations - Constructions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

09 / 24.4

Construct a perpendicular from outside

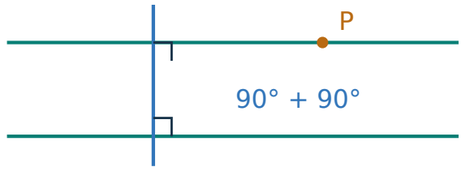


One circle centred at P cuts the line at A,B with PA=PB; a second equal-distance point Q fixes their bisector PQ.

Watch for: Drawing from P to just one crossing A instead of the matching-arc intersection.

10 / 24.4

Construct a parallel through a point

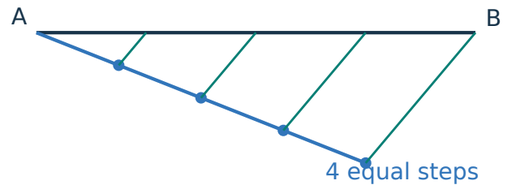


Two distinct coplanar lines perpendicular to the same line are parallel.

Watch for: Drawing a parallel-looking line that misses the specified point.

11 / 24.4

Divide a segment into equal parts

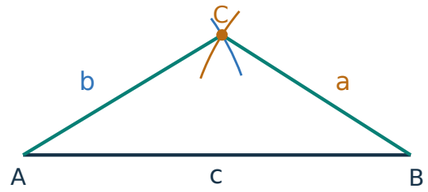


Equal steps on an auxiliary ray transfer to equal parts of AB through parallel lines; n is a positive integer.

Watch for: Using n−1 auxiliary steps for n required parts.

12 / 24.5

Construct a triangle from three sides

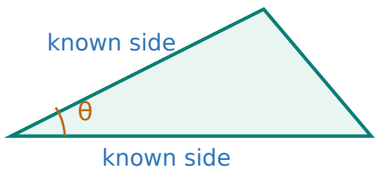


Each side must be positive; the two shorter lengths must sum to more than the longest.

Watch for: Using one radius for both arcs when AC and BC differ.

13 / 24.5

Construct a triangle from side-angle-side

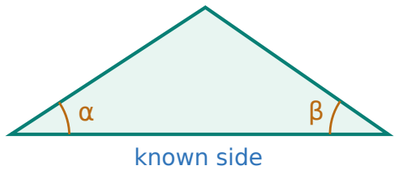


The angle lies between the two known sides; lengths>0 and 0°<included angle<180°.

Watch for: Placing the supplied angle opposite one given side instead of between them.

14 / 24.5

Construct a triangle from angle-side-angle

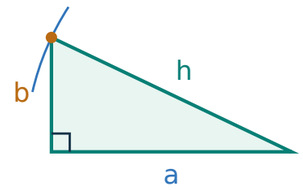


For endpoint angles α,β>0, require α+β<180°; the third angle is 180°−α−β.

Watch for: Measuring B’s angle from the outward continuation rather than BA.

15 / 24.5

Construct a right triangle from RHS

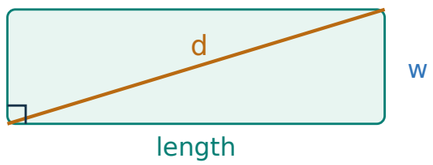


Hypotenuse h must exceed the given leg a>0; the other leg is √(h²−a²).

Watch for: Centring the hypotenuse arc at the right-angle vertex.

16 / 24.5

Construct and check a rectangle



Opposite sides equal and parallel; all angles 90°; diagonal²=length²+width².

Watch for: Using the diagonal as the rectangle’s width.